EEE/EIE Y2 Smart Grid Project

Group: Watt's Up

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Abstract

The Smart Grid is a DC microgrid built from five cooperating SMPS modules. It balances generation, storage and load on a shared 10V bus while minimising net grid cost through import, export and supercapacitor charge or discharge. Each converter regulates its own current flow against a local bus measurement, with a central coordinator setting dispatch commands. The system operates from a 12V primary supply onto a 10V bus with a ±0.1V droop band, allowing each module to act autonomously on the bus voltage signal. A Pico W on every module runs the control loop and feeds INA219 or SPI ADC readings to an operator dashboard. The dispatch algorithm schedules deferrable loads and supercapacitor cycles around price forecasts, beating the naive baseline by 20% in lab runs and by a mean of 28% across 1000 simulated days. A chassis printed in 3D houses the SMPS boards for airflow and mechanical protection.

Word count: ***9918***

Contents

1 Project Objectives & Management

1.1 Project Objectives

The project delivers a five-module DC microgrid that regulates a shared 10V bus across PV generation, supercapacitor storage, grid import, grid export and a programmable LED load. The control system dispatches power between modules to minimise net cost under a time-varying tariff. The prototype is evaluated on demo day, 18 June 2026, against a published price-and-load scenario.

1.2 Project Management System & Time Planning

Coordinating a seven-member hardware and software team across a five-week delivery window required tooling that off-the-shelf platforms could not provide. Rather than stitching together Notion, Trello, Microsoft Teams and ChatGPT, each of which solves only a fragment of the problem, an AI-powered Project Management System (PMS) was coded by our team. Every layer of the frontend, backend, AI pipeline and database schema was purpose built around our needs.

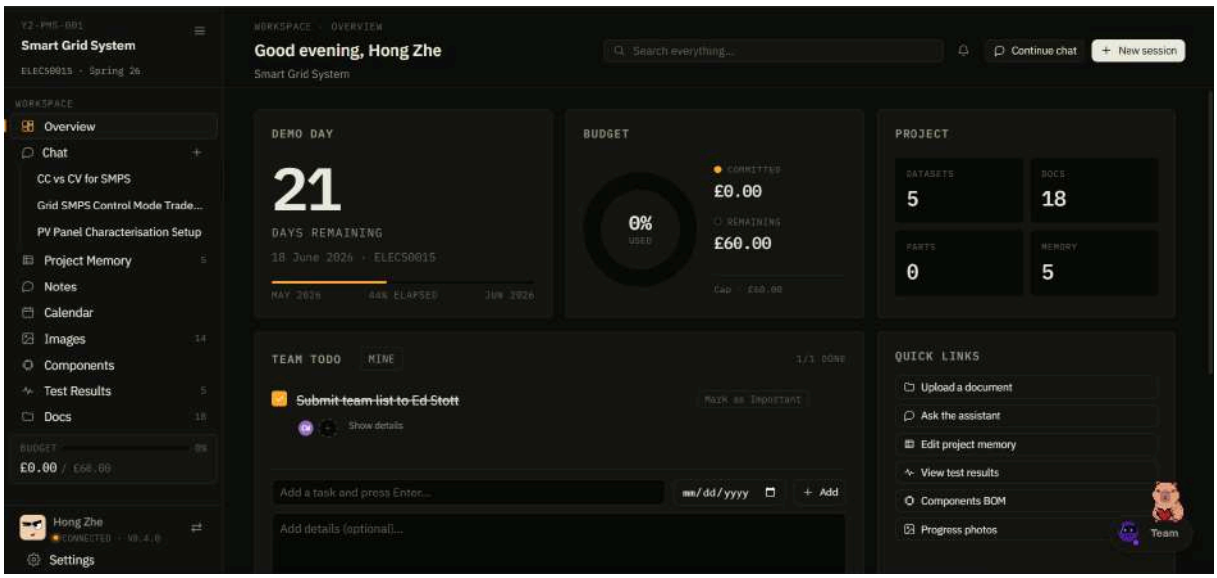


Figure 1.1: PMS dashboard: countdown, budget, stats and todo.

The platform is structured around four modules. A live operational dashboard (Figure 1.1) displays the project's current situation and deadlines. An automated meeting transcript module (Figure 1.2) records, transcribes and structures every meeting into attendee lists, discussion summaries and extracted action items. An AI-driven article and documentation generator named Aria (Figure 1.3) produces project-specific technical writing such as component-selection notes and lab write-ups. A Retrieval-Augmented Generation (RAG) project chat (Figure 1.4) compiles Aria articles, meeting minutes, the GitHub firmware repository, datasheets and test results into context for the AI to analyse and discuss with team members.

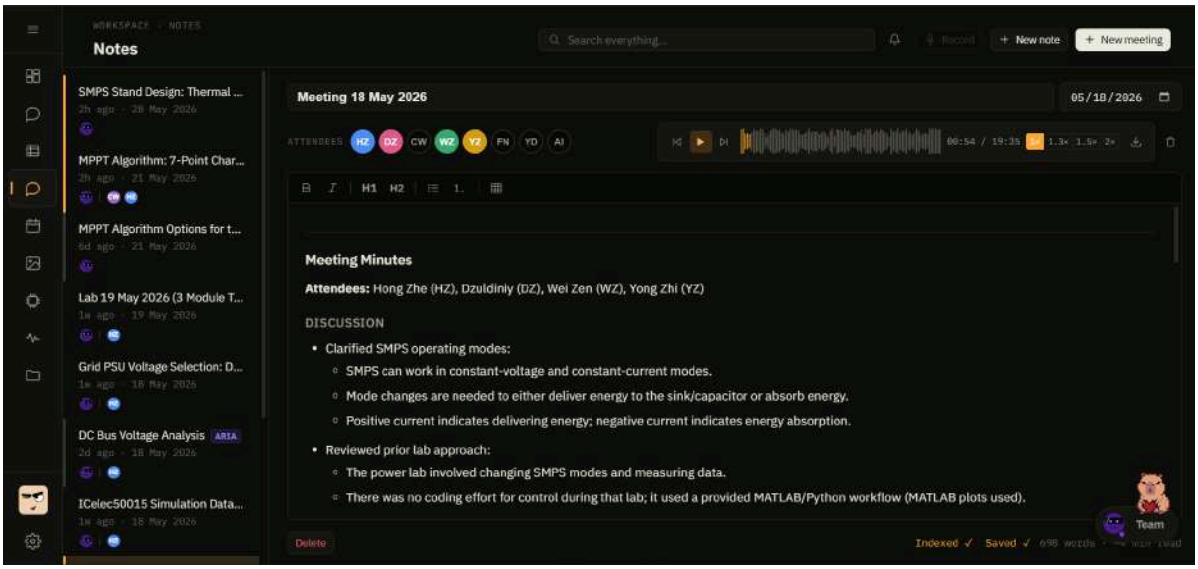


Figure 1.2: Meeting transcript module.

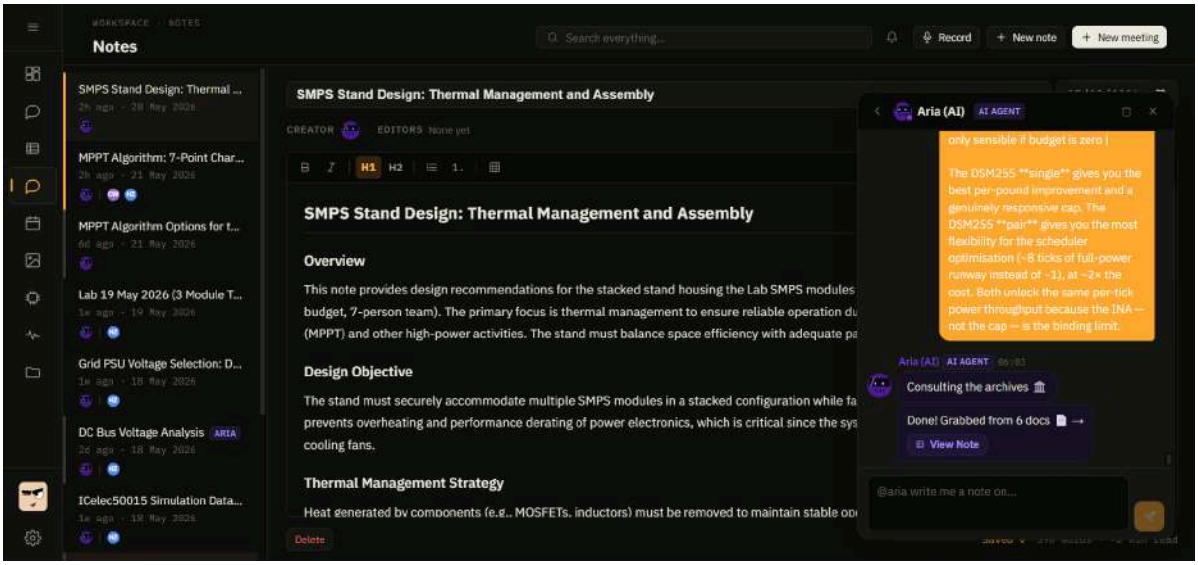


Figure 1.3: Aria article generator.

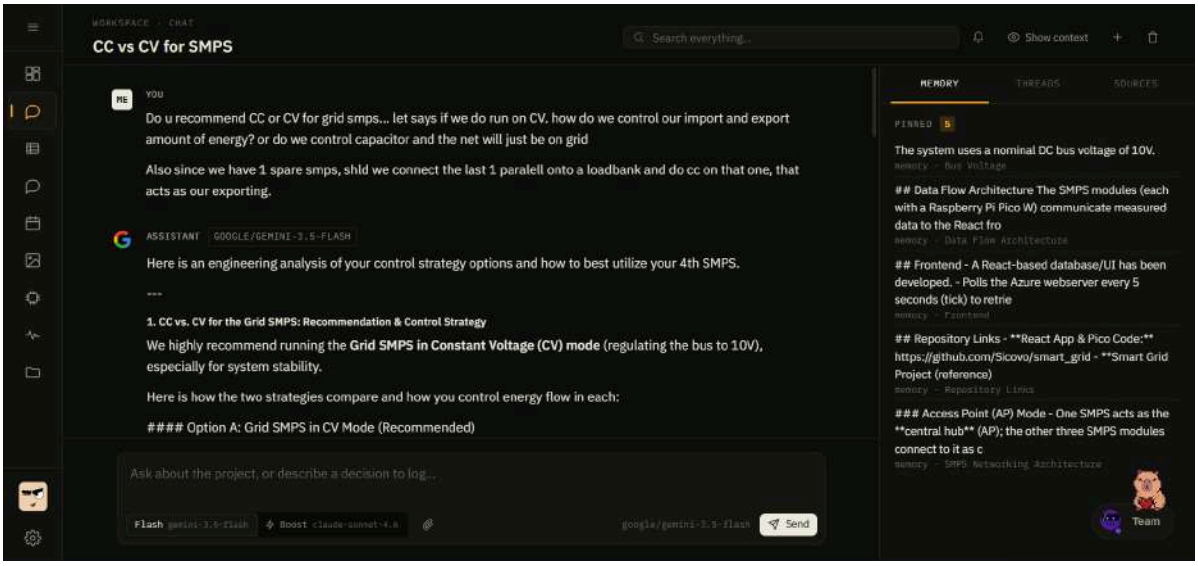


Figure 1.4: RAG project chat with cited sources.

Several additional features round out the platform (Figure 1.5): a Gantt-timeline calendar consolidating every milestone, a gallery centralising scattered lab photographs, a test-results module with built-in charting, and lightweight personalisation through a team buddy selector.

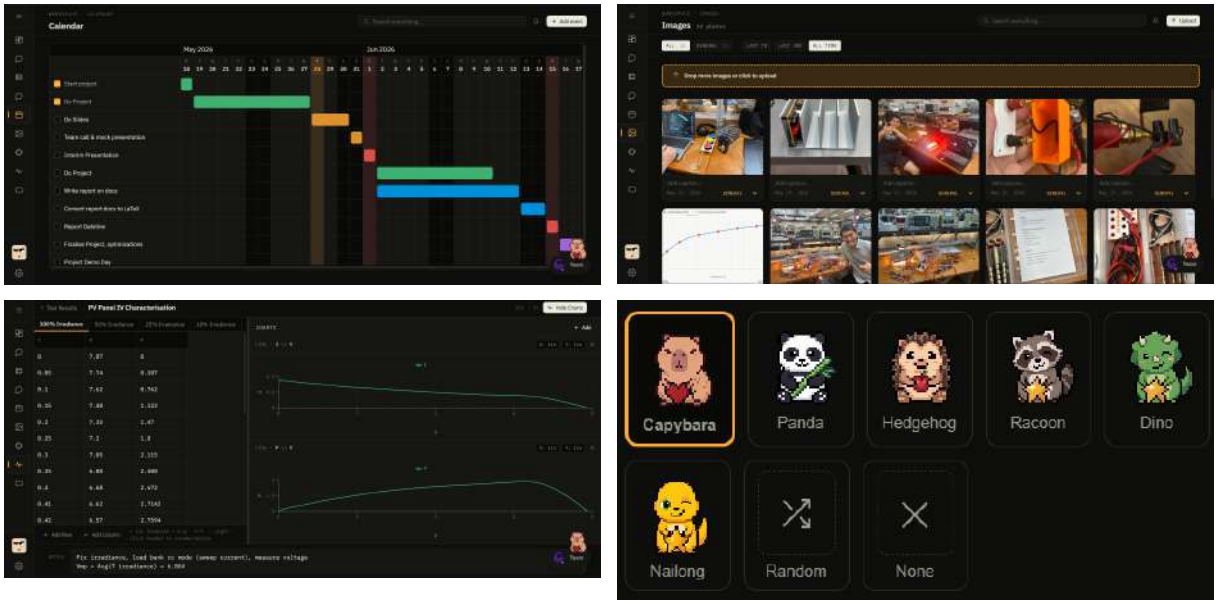


Figure 1.5: Calendar, image gallery, test results, and buddy selector.

1.3 Role Allocation

Work was split into one Task Master and one Task Assistant per hardware module and software stack, summarised in Table 1.1. The Task Master owned the design, build and report writing for their module, while the Task Assistant supported testing, ran cross-checks and reviewed the writing. Report-structure authoring was led by Hong Zhe, with Chun Wen and Yong Zhi acting as report checkers.

	PV Panel	LED Load	Import (PSU)	Export (Load Bank)	Supercapacitor	Backend	Frontend
Task Master	Chun Wen	Yong Zhi	Wei Zen	Yi Da	Hong Zhe	Dzuldiniy	Fangnan
Task Assistant	Hong Zhe	Yi Da	Chun Wen	Wei Zen	Yong Zhi	Fangnan	Dzuldiniy

Table 1.1: Role allocation for each hardware and software module.

2 System Architecture & Key Design Decisions

This section covers the hardware architecture: module roles, microgrid topology, bus and supply voltages, and the droop coordination scheme. The software architecture, dispatch algorithm and operator dashboard that sit on top are covered separately in Sections 8 and 9.

2.1 Module Roles

Module	External component	Topology	Control	Role on the bus
Grid Import SMPS	12V grid PSU	Buck	CV, droop at 9.9V	Sources current when bus < 9.9V
Grid Export SMPS	Sink resistor	Buck	CV, droop at 10.1V	Sinks current when bus > 10.1V
PV SMPS	PV panel	Boost	MPPT	Harvests panel power into bus
Supercapacitor SMPS	10.5 to 17.5V supercapacitor	Bidirectional buck/boost	CC, commanded	Stores or releases on command
LED Load SMPS	3V LED module	Buck (internal)	CC	Drives the LED load

Table 2.1: Topology, external component, control mode and bus role of each module.

The system comprises five SMPS modules connected to a shared 10V DC bus. The import and export modules share a synchronous-buck topology and a common droop, differing only in setpoints; this creates the dead-band discussed in Section 2.4. The Supercapacitor SMPS does not droop and runs as a commanded current source, so the economic algorithm in Section 9 can use it to buy energy cheap and sell it expensive. The LED Load SMPS is internally managed; only its bus-side draw is exposed. All four bidirectional modules use the same power stage in Figure 2.1, with a 100 µH inductor and a 100 mΩ shunt on Port B. The roles in the table depend on the port connections, not on the hardware, so each module's section refers back to this single stage.

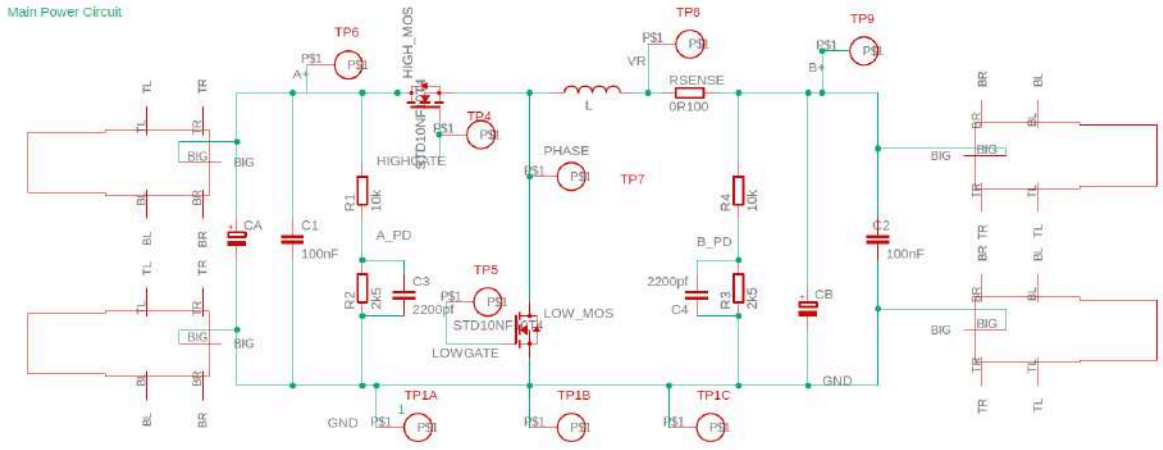


Figure 2.1: Bidirectional SMPS power stage, common to all four modules [1].

2.2 Overall Microgrid Topology

Figure 2.2 shows the complete system. External components sit at the top, each connected through its SMPS module to the shared 10V DC bus. A dedicated Pico W on every module runs the local control loop. A FastAPI backend polls the Azure web service [2] each second and, on a new tick, runs the dispatch decision (Section 9) and pushes setpoints to each module over HTTP (Section 8). The operator dashboard renders the live readings from the same backend.

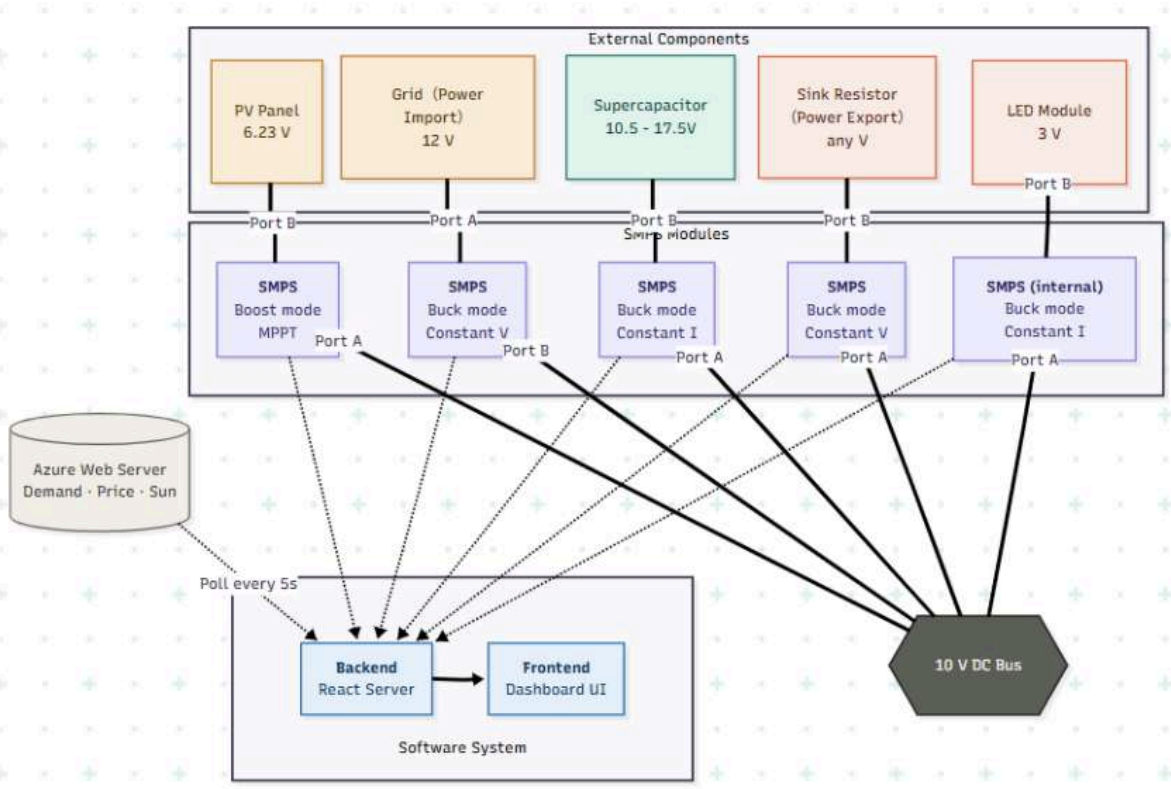


Figure 2.2: Smart Grid system architecture and power flow.

Every module is autonomous in its bus-regulation role, acting on its local V_{bus} measurement alone while the backend supplies only setpoints and economic context. All demo-day power measurements occur at the bus, never at SMPS terminals. The supercapacitor is the only element that stores energy for longer than the bus capacitance can, which puts its commanded behaviour at the centre of the economic optimisation.

2.3 Voltage Architecture: 10V Bus, ±0.1V Droop Band and 12V PSU

The voltage architecture is fixed by the operating windows of the PV panel, supercapacitor and grid-interface converters. A 10V nominal bus is the smallest round-number voltage that sits below the supercapacitor charge window, above the PV panel open-circuit voltage of about 8.5V, and reachable from a 12V buck input (justified below). Raising the bus would add switching stress with little benefit; lowering it would reduce the supercapacitor margin and increase the PV boost ratio.

Each droop-controlled module follows a local linear control law of the form

$$I_{\text{module}} = \frac{V_{\text{set}} - V_{\text{bus}}}{R_{\text{droop}}}$$

with sign and saturation set by module role. R_{droop} is sized so each module reaches its rated current at the edge of the ± 0.1V band, bounding the full-load steady-state bus error to $\pm 1\%$ of nominal. This keeps the bus stiff enough for downstream modules while regulation stays local.

The bus voltage choice also affects the scoring of converter losses. Demo-day measurements are taken at the bus, so LED Load and Export losses occur after the measured point and are counted as load or dissipated export power; their efficiency does not affect the score directly. Import losses occur before energy reaches the bus, so every watt lost between the 12V supply and the bus is grid energy missing from the measured balance.

The 12V PSU is the lowest practical input that preserves buck regulation with sufficient headroom, while avoiding the switching loss a higher input would add. For a synchronous buck converter [3],

$$P_{\text{cond}} = D \cdot I_{\text{rms}}^2 \cdot R_{\text{ds,on,HS}} + (1 - D) \cdot I_{\text{rms}}^2 \cdot R_{\text{ds,on,LS}}$$

which reduces to

$$P_{\text{cond}} = I_{\text{rms}}^2 \cdot R_{\text{ds,on}}$$

for matched MOSFETs. The dominant conduction term is therefore independent of both D and V_{in}. Switching loss scales with V_{in} or V_{in}^2:

$$P_{\text{sw}} \approx \frac{1}{2} \cdot V_{\text{in}} \cdot I_{\text{load}} \cdot \left(t_r + t_f\right) \cdot f_{\text{sw}} + \frac{1}{2} \cdot C_{\text{oss}} \cdot V_{\text{in}}^2 \cdot f_{\text{sw}}$$

and inductor ripple shrinks as V_{in} approaches V_{bus}:

$$\Delta I_L = (V_{\text{in}} - V_{\text{bus}}) \cdot D \cdot \frac{T}{L}$$

The binding constraint is dropout, not efficiency: with $D = \frac{V_{\text{bus}}}{V_{\text{in}}}$, duty tends to unity as the input approaches V_{bus}, but a real converter cannot reach $D = 1$ due to minimum off-time, gate-driver refresh and maximum-duty limits. With a maximum duty of about 0.9, regulation is lost below $\frac{V_{\text{bus}}}{0.9} \approx 11$V. A 12V bench supply leaves margin above this floor while staying close enough to the bus to limit switching loss.

Figure 2.3 shows a flat usable region rather than a strong efficiency optimum: across 11 to 15V at a 10V bus and 1.5A load, modelled loss changes by only tens of milliwatts, about 0.2% of delivered power. The sharp collapse below this region is dropout. The 12V point sits on the flat plateau with margin to spare.

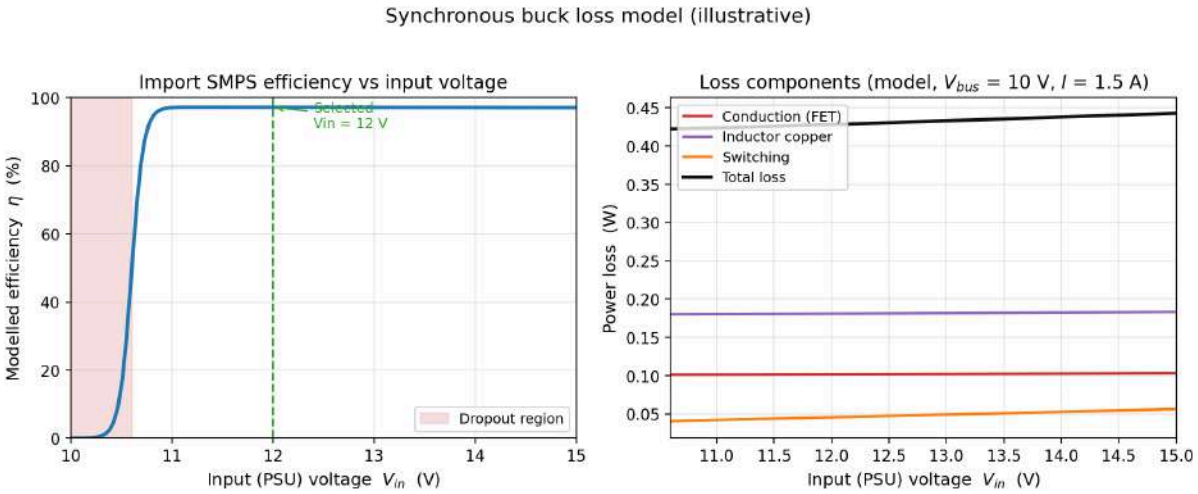


Figure 2.3: Modelled buck efficiency and loss split versus input voltage.

2.4 Coordination Scheme: Symmetric Droop vs Mutex Arbitration

Two architectures were considered. Mutex arbitration assigns bus control to one module at a time, with a coordinator switching modes as conditions change; symmetric droop, the established primary-control approach for DC microgrids [4], lets every module act at once on its local V_{bus} measurement. Mutex gives tighter steady-state regulation, but it places the communication channel inside the safety-critical bus-regulation loop (a dropped link or stalled coordinator collapses the bus), and every handover adds a voltage transient. Droop keeps each loop local and continuous: a communications failure degrades the economic optimisation but not bus stability, the curves blend without handover transients, and the cost is a small steady-state error bounded to ± 0.1V by design (Section 2.3).

Symmetry is enforced by mirroring the import and export setpoints about the 10V nominal:

$$V_{\text{set,import}} = 9.9 \text{ V} \qquad V_{\text{set,export}} = 10.1 \text{ V}$$

This creates a dead-band $V_{\text{bus}} \in [9.9, 10.1]$V in which neither import and export module acts and the bus is held by the PV module and supercapacitor alone, the no-cost configuration the algorithm in Section 9 maximises. Outside it, the system falls back to grid import or export with no explicit mode-change logic. Figure 2.4 and Figure 2.5 show the droop curves and a load cycle moving between import, dead-band and export regimes without central arbitration.

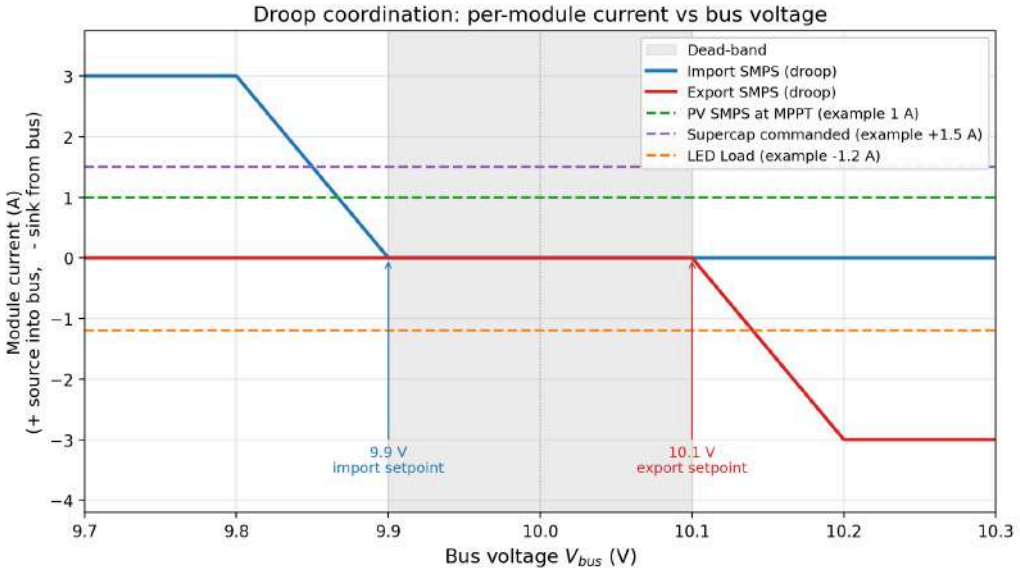


Figure 2.4: Droop curves: module current versus bus voltage.

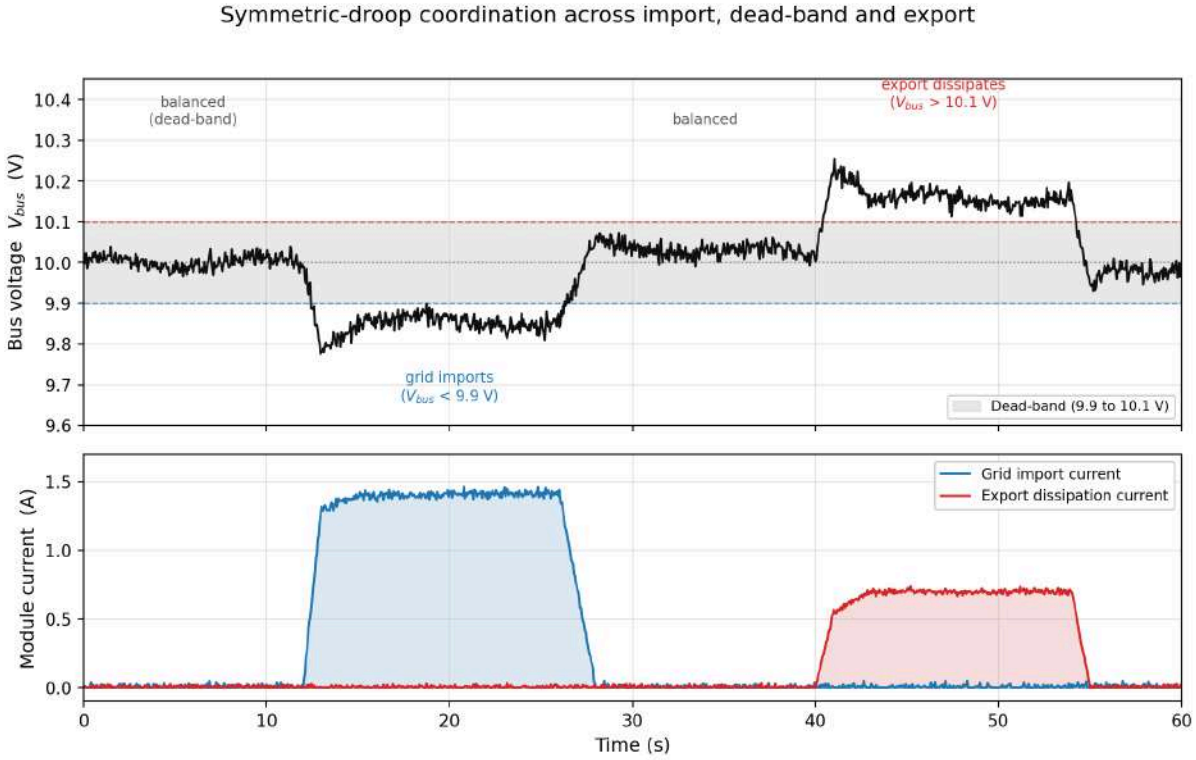


Figure 2.5: Bus voltage and converter currents across import, dead-band and export.

3 PV SMPS

3.1 Design & Simulation

3.1.1 Topology Selection

The bench setup uses one PV panel rated at approximately 4W at full illumination, with a datasheet open-circuit voltage of 8.5V and short-circuit current of 1.0A. Demo day adds a second identical panel in parallel for roughly double the available harvest. The single-panel characterisation in Section 3.3 transfers directly: each panel's MPP voltage is unchanged, so the MPPT lookup table, the P&O dwell behaviour and the boost converter's operating duty all stay valid; only the current at the bus doubles, which sits well under the 1.5A inductor current cap. Under the solar emulator the measured open-circuit voltage was about 7.5V, with the MPP near 5.20V at full irradiance and 5.10V at low irradiance (Figure 3.1). Each curve was taken by stepping the panel voltage from 2V toward the open-circuit knee (with denser sampling around the MPP) at one of six emulator dimmer settings (25% to 100%). Since the MPP voltage (around 5V) is below the 10V bus, energy reaches the bus by stepping up, so the module runs as a boost converter.

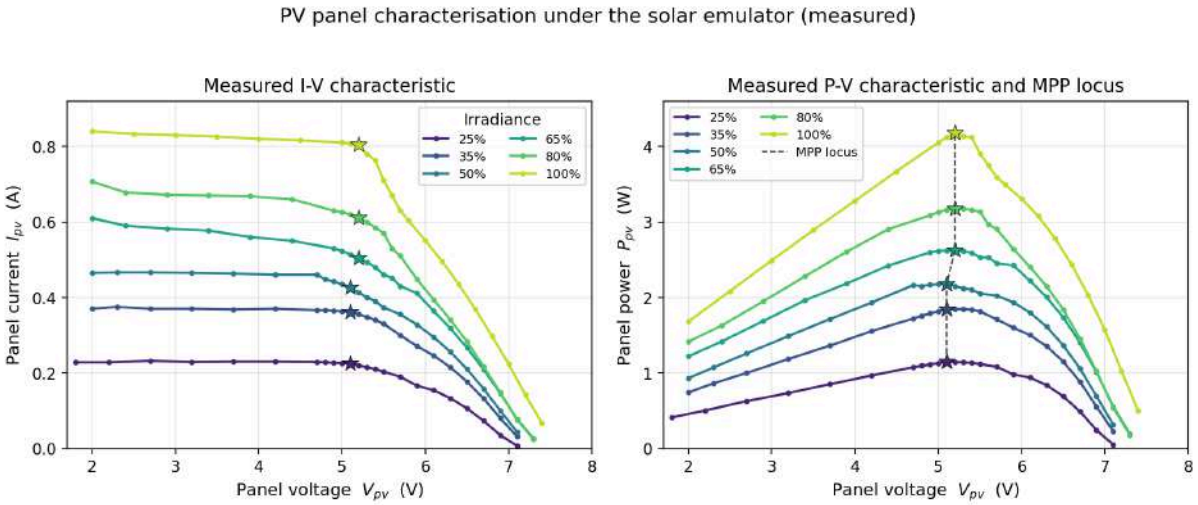


Figure 3.1: Measured PV panel I-V and P-V curves with the MPP points.

The panel is wired to Port B (lower, variable voltage) and the bus to Port A (fixed 10V), satisfying the hard constraint $V_A > V_B$. The BU/BO switch is set to BOOST, and unlike the buck modules the PWM is not inverted: the firmware writes duty directly rather than the `65536 - pwm_out` form.

Current is sensed by the INA219 across the 0.10 Ω Port B shunt. The shunt is wired in the Port A to Port B sense direction, so the firmware negates the reading; a positive inductor current then corresponds to panel-to-bus boost flow. The controller is cascaded: an outer voltage PI regulates the panel voltage to the commanded V_{mpp}, and an inner current PI drives the duty. The operating point is set by whichever MPPT strategy is active.

3.1.2 MPPT Strategy: Fixed Voltage vs Lookup Table vs P&O

Three MPPT approaches were prototyped: Fixed (constant V_{mpp}), an irradiance lookup table (LUT) from the Azure `/sun` feed, and online Perturb and Observe (P&O). Figure 3.2 shows V_{mpp} varying by only 0.10V across the full irradiance range, so a fixed 5.15V setpoint would lose

under 1% of available harvest at every characterised point at thermal equilibrium. The trade-offs are summarised in Table 3.1.

Mode	Description	Benefits	Disadvantages
Fixed	Constant V_{mpp} at 5.15V.	Simple; no external feed; no oscillation.	Open-loop on temperature.
LUT	V_{mpp} interpolated from the Azure `/sun` value.	Tracks irradiance; immediate settling.	Needs characterisation and the web link; open-loop on temperature.
P&O	Perturbs V_{mpp}, keeps or reverses by power comparison [5].	No external feed; corrects temperature drift.	Oscillates around the peak; noise-sensitive; slower convergence.

Table 3.1: MPPT mode comparison: Fixed, LUT and P&O.

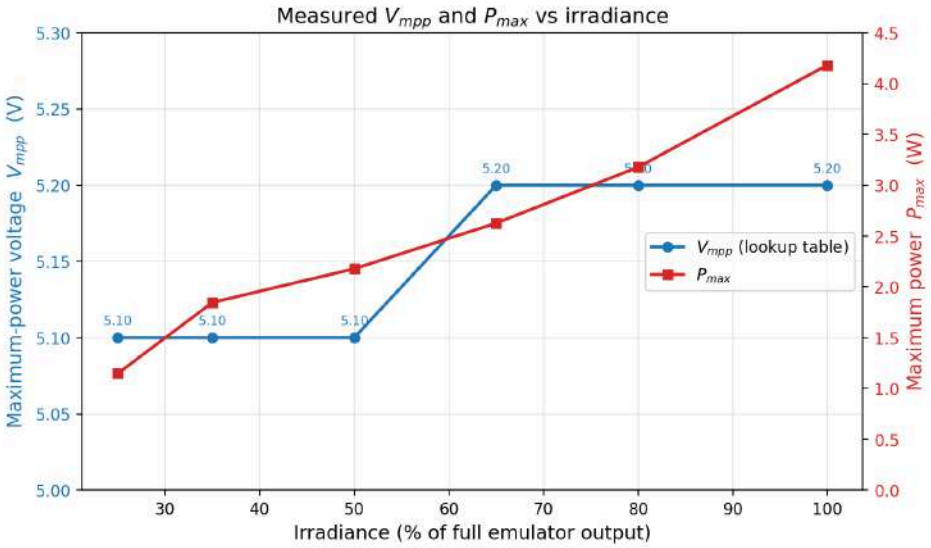


Figure 3.2: Measured Vmpp and Pmax versus irradiance: the MPPT lookup table.

Shipping all three gives a fair comparison under identical conditions and a fallback path if the web link fails: LUT drops to Fixed mode, and P&O stays available.

3.2 Testing & Implementation

3.2.1 Bench Noise Treatment

The three-layer noise treatment (Section 7) was applied to the PV readings before testing began; the filtered values feed the dashboard and the P&O power comparison while raw readings are retained for the control loop and trip guards.

3.2.2 P&O Implementation

P&O uses a two-timescale structure. The outer voltage PI tracks the commanded V_{mpp} at 100 Hz; perturbations occur at a 300 ms interval (T_{dwell}), roughly five times the outer-loop settling time, so the power measured at the end of a dwell reflects steady state, not the PI transient. The comparison power is the filtered $V_{\text{panel,lp}} \cdot I_{\text{L,lp}}$ accumulated over the second half of each dwell and averaged. Two-stage filtering (EMA then dwell average) brings the noise floor of each comparison under 10 mW, well below the $P(V_{\text{mpp}})$ curvature near the peak.

A direction-flip rule with a dead-band stops the three-point oscillation degenerating into noise. If dwell-averaged power dropped by more than $\varepsilon = 20$ mW versus the previous dwell, direction reverses; otherwise it continues. The commanded V_{mpp} is hard-clamped to 4.75V to 5.5V, a narrow window centred on the measured 5.10 to 5.20V MPP range (Figure 3.2), so an algorithm fault cannot push the operating point into open-circuit or short-circuit territory. On entry to P&O mode the state is reseeded from the last commanded fixed-mode V_{mpp}, avoiding a stale value. A three-second block average of P_{panel} (restarting on each mode switch) lets the operator read steady-state power for each mode from the dashboard, separate from the live tracking-noise trace.

3.3 Evaluation & Optimisation

3.3.1 MPPT Mode Comparison

Each MPPT mode is tested at 25%, 50%, 75% and 100% irradiance. At each setting the three-second block-averaged P_{panel} is recorded in fixed mode, LUT mode, and P&O mode. $V_{\text{mpp,target}}$ and v_{panel} are logged throughout so the mode-switch transient can be inspected. The primary metric is steady-state delivered power; the secondary metric is the mode-change transient.

Mean recorded powers per mode and irradiance are summarised in Figure 3.3 and tabulated below.

Irradiance	Fixed (W)	LUT (W)	P&O (W)	Spread (mW)
25%	0.767	0.770	0.770	3
50%	1.933	1.932	1.933	1
75%	2.845	2.841	2.841	4
100%	3.577	3.562	3.550	27

Table 3.2: Recording-phase mean panel power per MPPT mode and irradiance.

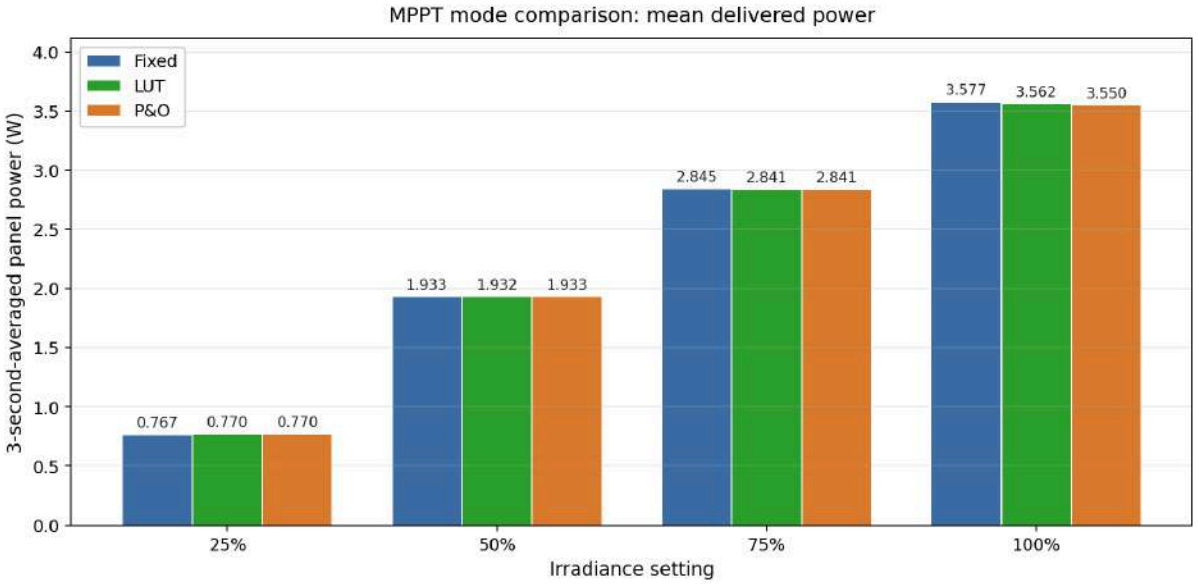


Figure 3.3: Mean delivered panel power by MPPT mode at four irradiance settings.

The three modes converge to within a few mW at every irradiance; the worst-case spread of 27 mW at 100% is under 1% of the 3.55 W operating point. This is what the construction predicts: P&O hunts to the same peak the LUT is built from, and Fixed sits within 0.10V of V_{mpp} across

the full range (Figure 3.2), so all three settle close to the same operating point, with no steady-state harvest advantage at thermal equilibrium under the bench emulator.

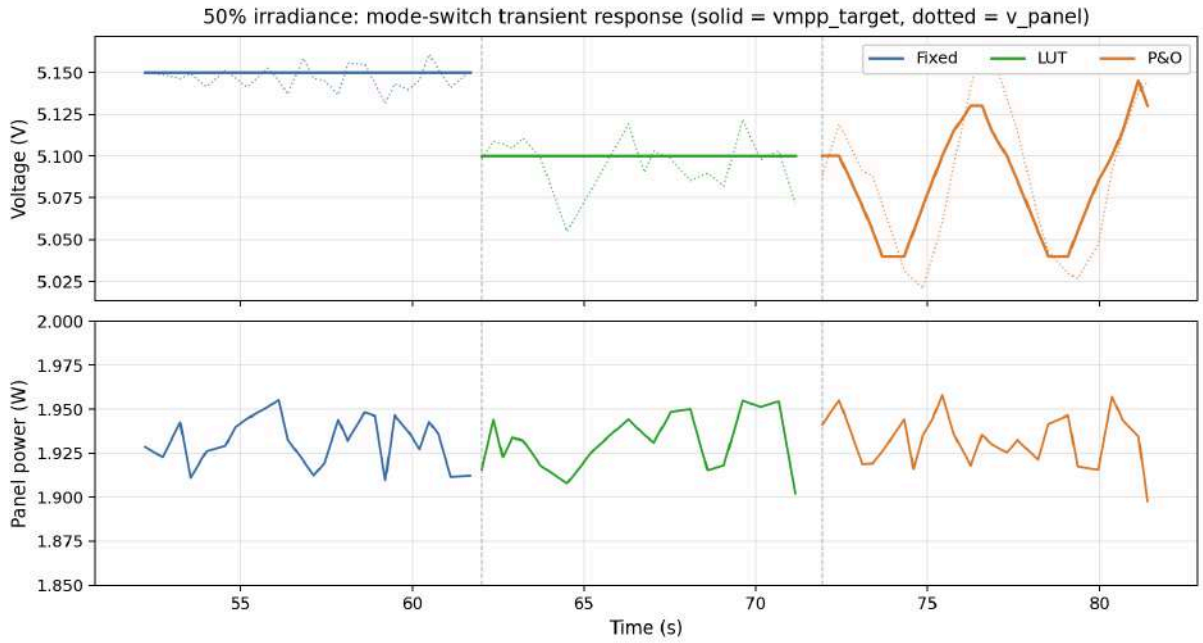


Figure 3.4: Mode-switch transient at 50% irradiance: commanded V_{mpp} (solid), measured v_{panel} (dotted), and resulting P_{panel}.

Figure 3.4 shows the mode-switch transient at 50% irradiance. Fixed and LUT hold their commanded V_{mpp} flat at 5.15V and 5.10V, with v_{panel} tracking inside the noise envelope. P&O shows the expected three-point hunting oscillation around the peak, stepping V_{mpp} up and down in a bounded cycle; the resulting panel power stays inside the same envelope as the other two modes, confirming the algorithm locks onto the peak rather than drifting.

The real differentiator is temperature: as the panel warms under sustained illumination, V_{mpp} drifts downward by several mV/°C [6]. Fixed and LUT cannot follow this since both are open-loop, whereas P&O corrects automatically by optimising power directly. P&O is therefore the shipped default, with Fixed and LUT kept as fallbacks in case of a fault.

3.3.2 Measured SMPS Efficiency

PV conversion efficiency is defined as $\eta_{\text{PV}} = \frac{P_{\text{bus,out}}}{P_{\text{panel,in}}}$, with $P_{\text{panel,in}} = V_{\text{panel}} \cdot I_{\text{L}}$ from firmware and $P_{\text{bus,out}}$ measured by a DMM in series with the PV-to-bus wire. Figure 3.5 and Table 3.3 show the 48-point sweep: four irradiance settings at twelve bus-voltage points each, with the panel pinned at its MPP ($V_{\text{panel}} \approx 5.15$ V).

V_{bus} (V)	Duty D	η_{PV} at 25%	η_{PV} at 50%	η_{PV} at 75%	η_{PV} at 100%
7.11	0.28	96.56%	96.69%	97.17%	97.68%
7.61	0.32	96.36%	96.79%	96.82%	97.89%
8.11	0.37	96.62%	96.57%	96.98%	97.74%
8.61	0.40	96.52%	96.44%	96.73%	98.00%
9.12	0.44	96.16%	96.63%	96.79%	97.86%
9.61	0.46	96.69%	96.34%	97.11%	98.10%
10.21	0.50	96.56%	96.80%	97.15%	98.02%
10.63	0.52	96.57%	96.68%	97.28%	97.79%
11.07	0.54	96.18%	96.75%	96.90%	98.10%
11.63	0.56	96.37%	96.51%	96.92%	98.08%
12.06	0.57	96.32%	96.88%	96.98%	97.83%
12.55	0.59	96.66%	96.84%	96.81%	97.77%

Table 3.3: Measured PV SMPS efficiency across bus voltage and irradiance.

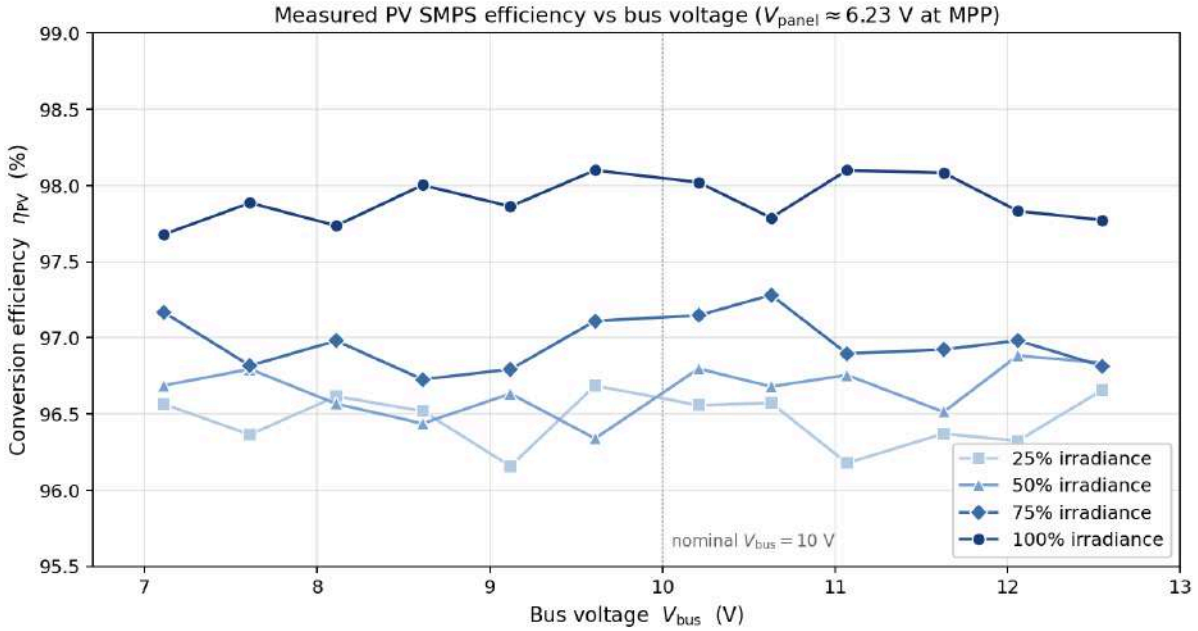


Figure 3.5: Measured PV SMPS efficiency versus bus voltage at four irradiances.

Each η_{PV} trace is flat across bus voltage: the four curves vary by no more than $\pm 0.3\%$, consistent with the boost η-versus-D plateau over $D = 0.28$ to 0.59 (Figure 3.6 [7]). The one exception is the supercapacitor-full corner ($V_{\text{bus}} = 17.5$ V, $D_{\text{max}} \approx 0.71$), which sits at the CCM efficiency cliff where degraded efficiency would be expected.

Mean efficiency rises from 96.5% at 25% irradiance to 97.9% at 100%, a 1.4 percentage-point spread consistent with fixed switching overhead consuming a larger fraction of the lower throughput at reduced irradiance.

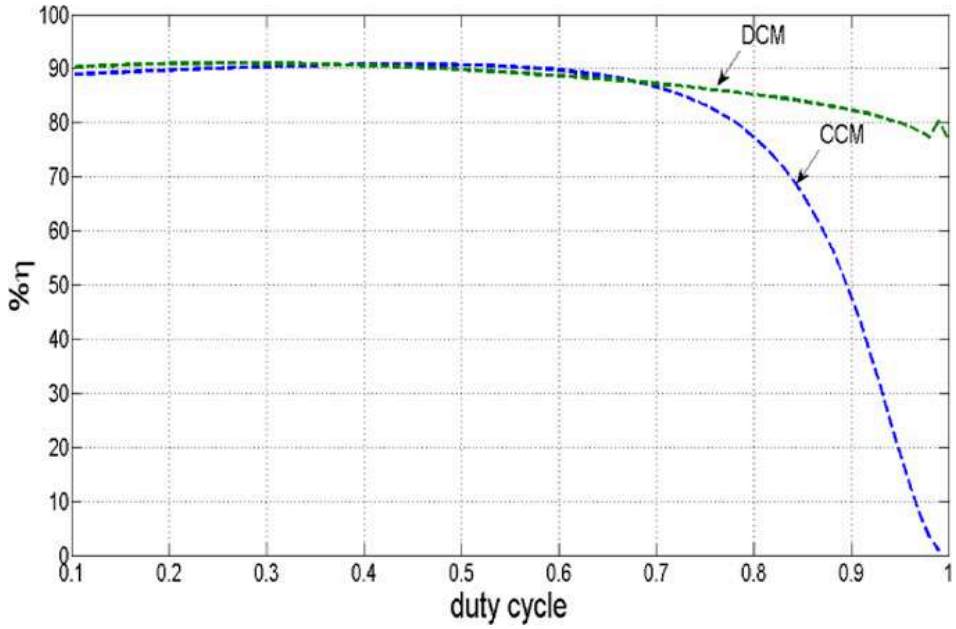


Figure 3.6: Boost efficiency versus duty cycle in DCM and CCM, from Fig. 5 of [7].

Import efficiency does enter the score under the dual-meter scoring of Section 2.3 and rises steadily with V_{bus} (Section 5.3), so the import lens marginally favours a higher bus voltage; the PV lens is neutral. The $V_{\text{bus}} = 10$V choice rests on hardware constraints (supercapacitor lower bound, PV panel open-circuit voltage, buck dropout), with both bus-relevant efficiency curves consistent with that pick.

4 Supercapacitor SMPS

4.1 Design & Simulation

4.1.1 Boost Topology and Boot-with-Empty-Capacitor Rationale

The supercapacitor is the only variable-voltage element on the storage side, ranging from 10.5V when depleted to 17.5V when full. The same Port A above Port B rule that fixed the PV topology applies here (Figure 2.1): the capacitor is the higher, variable rail on Port A, the fixed 10V bus on Port B. Energy flows bus-to-capacitor (B to A) when charging and back (A to B) when discharging, so the module is bidirectional.

The SMPS is set to BOOST mode, the key robustness choice. In BOOST mode, the on-board circuitry draws from Port B, the bus, which is always energised by the grid and PV modules. As a result the SMPS boots and stays alive even when the capacitor is empty at 0V. The alternative BUCK setting would power the bus from Port A, the capacitor, and would trip the SMPS whenever the capacitor fell below the 6V to 7V minimum support voltage. Following the PV sign convention, positive inductor current means charging, negative means discharging, and i_{cmd} is signed accordingly.

4.1.2 Operating Window 10.5 to 17.5 V

The usable voltage window is determined by two hard limits, illustrated in the stored-energy curve in Figure 4.1. The ceiling $V_{\text{cap,max}} = 17.5\text{V}$ is the SMPS port limit and sits under the 18V capacitor rating. The floor $V_{\text{cap,min}} = 10.5\text{V}$ provides a 0.5V margin above the 10V bus to maintain $V_{\text{A}} > V_{\text{B}}$ under ESR-induced voltage drop and measurement noise. Within this window the capacitor stores energy as $E = \frac{1}{2}CV^2$ with $C = 0.5\text{F}$ (two 0.25F capacitors in parallel).

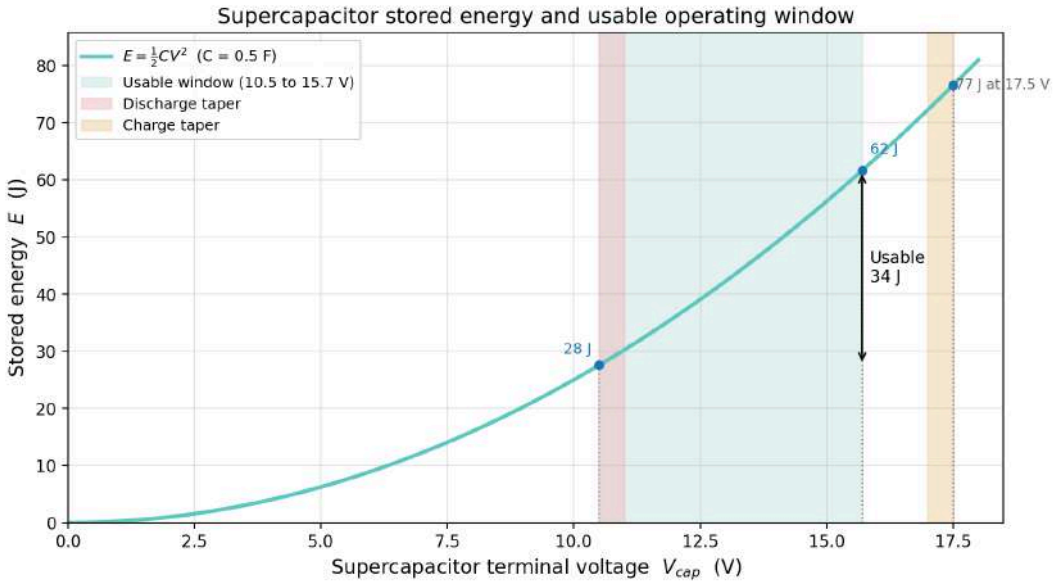


Figure 4.1: Supercapacitor stored energy and usable operating window.

In firmware the practical voltage ceiling is lowered to 15.7V, to reduce the impact of noise overshooting the voltage limit. The current is reset to 0 whenever the capacitor approaches within 0.5V of the upper and lower limits. The energy genuinely transferred between 10.5V and 15.7V is approximately 34J of the 49J the full window could hold. A significant complication is the capacitor's approximately 4 Ω equivalent series resistance (ESR): after a current step the

terminal voltage is dominated by resistive drop and charge redistribution, so a true reading needs a settling pause. This shapes the cycle-test settle phases (Section 4.2), which read the open-circuit voltage only after a deliberate pause. Peak current is clamped to 0.60A, leaving approximately 21% headroom under the 0.76A SMPS hardware limit [8].

4.1.3 Command-Driven CC: Rationale for the Departure

Unlike the grid and export modules, the supercapacitor is not involved in droop. The bus voltage is already regulated by the grid and export droop pair (Section 2.4). Instead the capacitor is a commanded current source and sink: it tracks a signed current given by input from the economic algorithm. This allows arbitrage: charging when energy is cheap and discharging when it is expensive, rather than reacting passively to instantaneous bus voltage.

The control law is an inner current PI only. The current limits implemented are a maximum or minimum at ± 0.60 A, a linear taper to 0 near the voltage limits, and a bus-health taper that reduces charging demand when the bus voltage sags, so the capacitor does not compete with the grid import module for current during a shared deficit.

4.2 Testing & Optimisation

The supercapacitor evaluation answers two questions: what is the round-trip energy efficiency η_{RT}, and how does it split into η_{charge} and $\eta_{\mathrm{discharge}}$? Both are answered from a single charge-and-discharge cycle, using the capacitor's known capacitance as an internal calibration reference. The same dataset supplies the efficiency figures the dispatch algorithm in Section 9 relies on.

4.2.1 Automated Cycle-Test Method

For a complete cycle from V_{low} to V_{high} and back, the firmware integrates bus-side input energy during charge,

$$E_{\mathrm{in}} = \int V_{\mathrm{bus}} \cdot I_{\mathrm{bus}} dt$$

and bus-side output energy during discharge,

$$E_{\mathrm{out}} = \int V_{\mathrm{bus}} \cdot |I_{\mathrm{bus}}| \; dt$$

The round-trip efficiency is then

$$\eta_{\mathrm{RT}} = \frac{E_{\mathrm{out}}}{E_{\mathrm{in}}}$$

This metric is preferred because both E_{in} and E_{out} are measured at the bus, so the result does not depend on estimating capacitor-side current. The capacitor-side stored energy $E_{\mathrm{cap}} = \frac{1}{2} C V_{\mathrm{cap}}^2$ is still recorded before charge, after charge, after any hold phase, and after discharge. It is useful for checking the energy swing but not as the main metric, since the terminal voltage relaxes after current steps (ESR and charge redistribution).

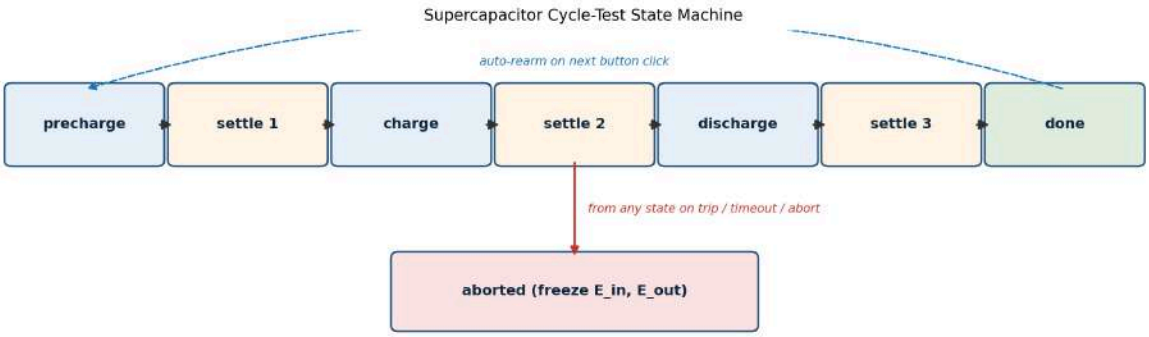


Figure 4.2: Supercapacitor cycle-test state machine.

As shown in Figure 4.2, the test is implemented as a firmware state machine triggered from the dashboard. It runs through precharge, settle 1, charge, settle 2, optional hold, discharge, settle 3 and done. Each settle phase records the open-circuit capacitor voltage only after a short delay, reducing the effect of the immediate ESR transient. Bus-side energy integration runs at the 1 kHz control tick, so the reported η_{RT} is not limited by the dashboard logging rate. The discharge endpoint is kept at $V_{\mathrm{low}} = 10.5\mathrm{V}$, matching the usable operating window, so any taper near the lower limit is included in the measured round-trip result.

Direct separation of η_{charge} and $\eta_{\mathrm{discharge}}$ from capacitor-side voltage was unreliable: ESR-driven redistribution caused the discharge-side estimate to exceed 100% in several runs. The reliable figure carried forward is the bus-side round-trip efficiency η_{RT}.

4.2.2 Round-Trip Efficiency Results

Two no-hold sweeps were run, with each operating point repeated three times. The first varied charge and discharge current from 0.1A to 0.6A at a 10V bus. The second varied bus voltage from 8.0V to 10.0V at 0.2A. Figure 4.3 shows a representative 10V, 0.2A cycle: the capacitor charges from 10.5V to the 15.7V cutoff, then discharges back to 10.5V. The measured bus-side energies are $E_{\mathrm{in}} = 38.2$ J and $E_{\mathrm{out}} = 33.7$ J, giving $\eta_{\mathrm{RT}} \approx 88\%$. The capacitor-side energy swing is approximately 34J, matching the design estimate from the usable voltage window.

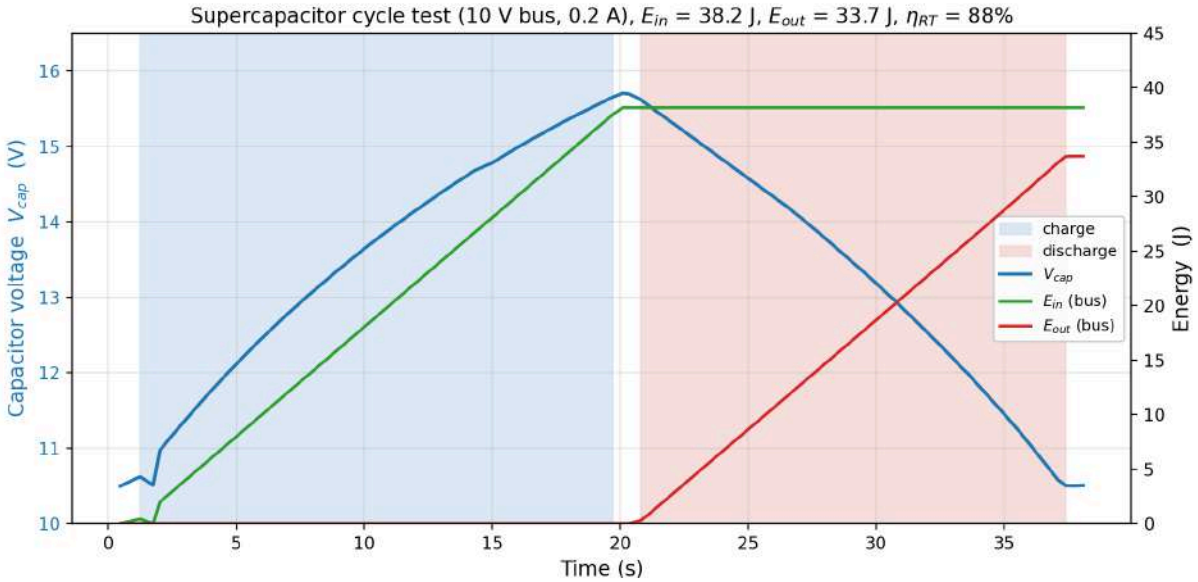


Figure 4.3: Measured supercapacitor charge and discharge cycle at 10V, 0.2A.

The measured round-trip efficiency is stable across the tested range. In the current sweep, η_{RT} falls only slightly from 89.7% at 0.1A to 86.9% at 0.5A, with the 0.6A result recovering to 87.4% within run-to-run scatter. This weak downward trend is consistent with conduction loss increasing with current. In the bus-voltage sweep, efficiency varies even less, rising mildly from 87.3% at 8.0V to 88.3% at 10.0V. This is consistent with the smaller boost ratio when the bus voltage is closer to the capacitor voltage.

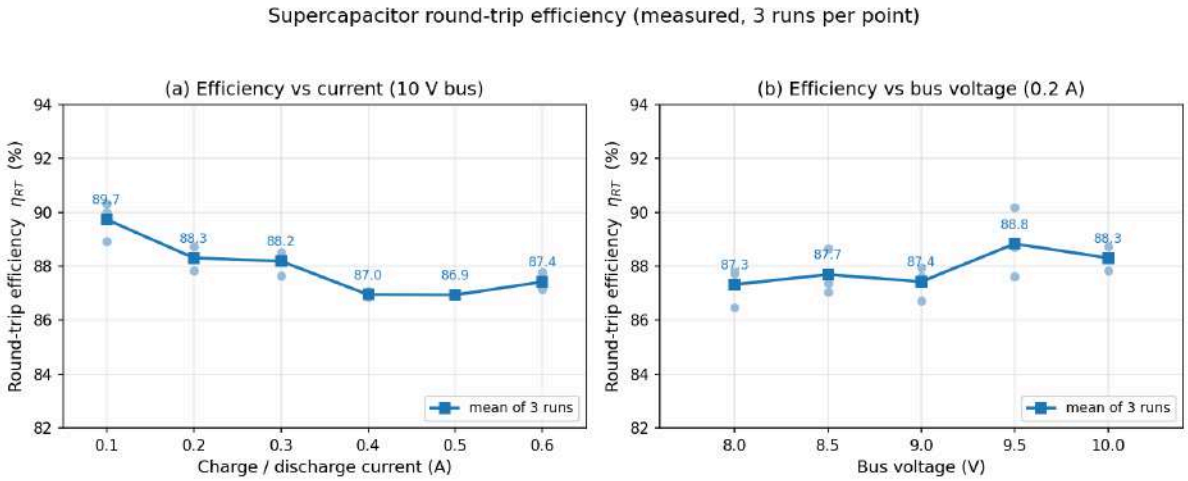


Figure 4.4: Measured round-trip efficiency versus current and versus bus voltage.

Charge current	Mean η_{RT}, 3 runs
0.1 A	89.7%
0.2 A	88.3%
0.3 A	88.2%
0.4 A	87.0%
0.5 A	86.9%
0.6 A	87.4%

Table 4.1: Supercapacitor round-trip efficiency by charge current at 10 V bus.

Bus voltage	Mean η_{RT}, 3 runs
8.0 V	87.3%
8.5 V	87.7%
9.0 V	87.4%
9.5 V	88.8%
10.0 V	88.3%

Table 4.2: Supercapacitor round-trip efficiency by bus voltage at 0.2 A.

The supercapacitor path therefore achieves approximately 88% immediate round-trip efficiency across the intended operating range. This figure is higher than the conservative 70 to 85% pre-test estimate and is sufficiently stable to use in the system-level dispatch model.

4.2.3 Self-Discharge and Hold-Duration Dependence

The same cycle test was extended with a zero-current hold phase between charge and discharge. Hold durations of 60, 120, 180, 240 and 300 seconds were tested at 0.2A and a 10V bus, with three repeats per point. This isolates the effect of storing energy in the capacitor before using it.

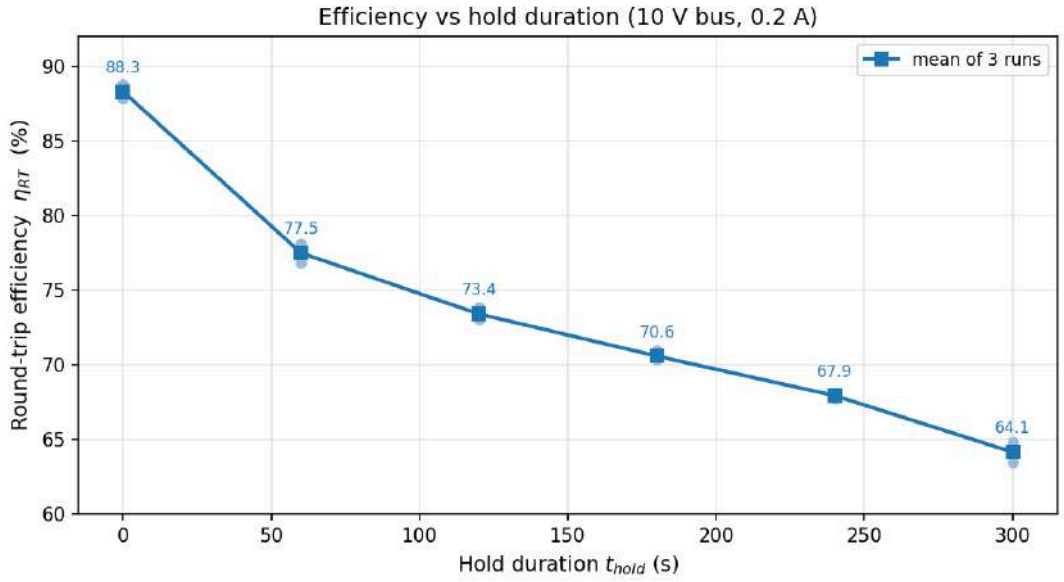


Figure 4.5: Round-trip efficiency versus zero-current hold duration.

Round-trip efficiency falls sharply with hold time: from 88.3% with no hold to 77.5% after 60 seconds, and to 64.1% after 300 seconds. Because $\eta_{\mathrm{RT}} = \frac{E_{\mathrm{out}}}{E_{\mathrm{in}}}$ is measured entirely from bus-side energy, this drop is a real loss of recoverable energy rather than an error in the capacitor-voltage measurement.

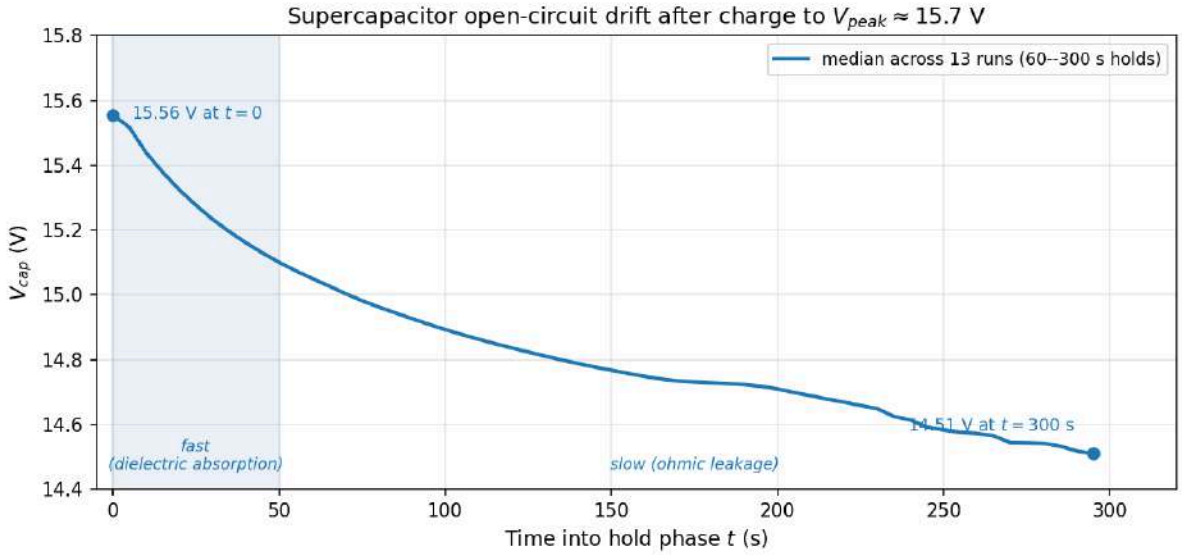


Figure 4.6: Open-circuit V_{cap} drift after charge: master curve across 13 hold runs.

Also as seen in Figure 4.6, after charging to the 15.7V cutoff, V_{cap} falls quickly by about 0.5V over the first 50 seconds, then continues to decay more slowly at roughly 2 to 3 mV/s. The fast component is attributed to charge redistribution within the supercapacitor [9], while the

slower component is consistent with leakage. Since all hold-duration runs collapse onto the same relaxation curve, the loss can be treated as a repeatable function of hold time.

For dispatch this means the supercapacitor should be charged just-in-time rather than charged early and held near peak voltage. The economic algorithm should therefore delay charging until the stored energy is actually needed, unless the alternative is wasting PV energy that cannot otherwise be used.

5 Import & Export SMPS

5.1 Design & Simulation

5.1.1 Topology Choice

Both import and export modules are configured as synchronous buck converters with the BU/BO switch set to BUCK (the shared power stage is Figure 2.1). The import module places the 12V bench PSU on Port A and the 10V bus on Port B, sourcing current into the bus. The export module places the 10V bus on Port A and a resistor bank on Port B, stepping the voltage down across the resistor to dissipate energy. Current cannot be pushed into the PSU or bank, so the minimum reference current for both SMPS is 0. This satisfies the Port A above Port B rule.

The duty cycle is inverted for these SMPS compared to the PV and supercapacitor boost SMPS due to the inversion of the gate driver. Current sensing is performed by the INA219 driver across the 0.1 Ω Port B shunt. The choice of a dummy resistor is in accordance with the project brief [10]: using a resistor sized to sink the maximum export current instead of driving current into a bench PSU, which would collapse the terminal voltage.

The maximum export power is the sum of maximum PV power and capacitor power. The demo-day configuration of two parallel PV panels supplies a maximum of roughly 8W at the panel side, and the capacitor supplies a maximum of about 13W at peak 0.6A discharging current near the top of its window, making the maximum export power under 20W. With 10V at the input, the 20W load needs $\frac{20}{10} = 2$ A through a $\frac{10}{2} = 5\ \Omega$ bank. The inductor-current reference is therefore capped at 2 A (the 20W corner), under the module's 2.8A firmware overcurrent trip.

5.1.2 Droop Setpoints: 9.9 V Import and 10.1 V Export

Each module runs a cascaded controller: an outer bus-voltage PI produces a reference current and an inner current PI produces a PWM which changes the voltage (Figure 5.1). The two modules differ only in the sign of the voltage error. The import module computes $v_{\text{err}} = V_{\text{target}} - V_{\text{bus}}$, so it sources more current the further the bus falls below its 9.9V setpoint. The export module computes $v_{\text{err}} = V_{\text{bus}} - V_{\text{target}}$, the sign flipped, so it dissipates more the further the bus rises above its 10.1V setpoint. Each reference current is limited to 2A on both modules (approximately 20W at the bus), the export value set by the worst-case surplus analysis in Section 5.1.1 and the import value by the grid interface rating.

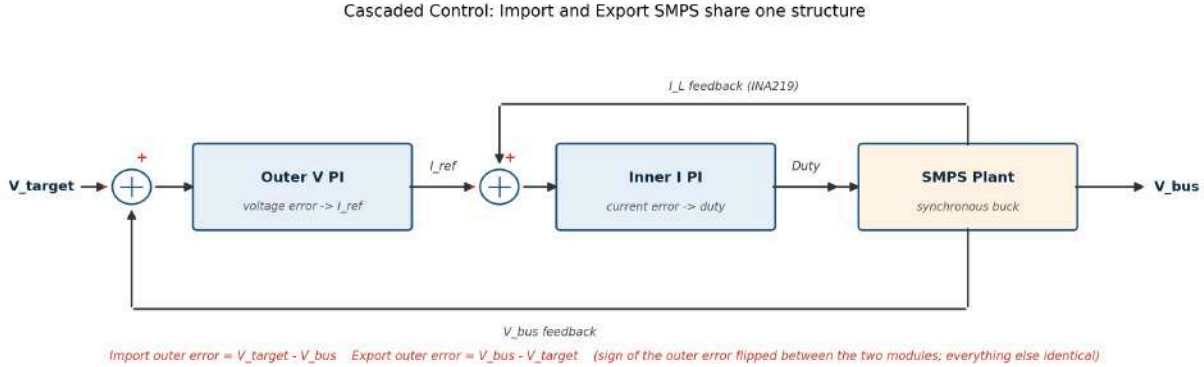


Figure 5.1: Cascaded control structure shared by Import and Export SMPS modules.

The dead-band that the symmetric setpoints create is analysed in Section 2.4. For implementation, a setpoint of 0.1V or below, or a disable command, is treated as fully off, which is how the bus changes from entirely import to entirely export.

5.2 Testing & Implementation

Both bus-facing modules received the same three-layer noise reduction as the PV module (Section 3.2), applied through the shared `INA219` class so import and export pick it up in one edit.

5.2.1 Anti-Windup and Soft-Start Grace

The outer voltage PI on each module uses clamping anti-windup, with the integrator bounded to the range that just saturates the current reference. At the default bound the integrator wound up far past saturation and took several seconds to unwind, observed as the rail fluctuating between 8V and 12V at about 1Hz when the PV and export SMPS ran together without the grid SMPS.

A 500ms soft-start grace disables the undervolt trip on every transition from inactive to active (enable, watchdog clear and trip reset), since the bus starts near 0V and would otherwise trip immediately; overvolt and overcurrent trips stay enabled throughout. The export module carries the same grace logic, with its output current clamped to 0 while the bus voltage is below target.

5.3 Evaluation & Optimisation

Under the dual-meter scoring of Section 2.3, import conversion loss costs money directly: the PSU reading is the billed quantity, so every 1% of loss inflates the effective import price per delivered bus Wh. Export loss instead appears as a gap between the dashboard's post-loss dissipation and the bus meter's pre-loss credit (Section 6.3.2 covers the same effect on the LED path; the fix in both cases is a current sensor between the bus and the SMPS). The import sweep is the characterisation that matters for cost.

The import SMPS efficiency is $\eta_{\text{import}} = \frac{P_{\text{bus,delivered}}}{P_{\text{PSU,drawn}}}$. The pre-loss draw $P_{\text{PSU,drawn}} = V_{\text{PSU}} \cdot I_{\text{PSU}}$ is not measured on the import module because the shunt sits on Port B. Unlike the PV case in Section 3.3, I_{PSU} can be read from the bench PSU's built-in current display with manufacturer-stated accuracy (the same display the lecturer reads on demo day).

The bus-voltage axis is swept across nine points from 7V to 11V in roughly 0.5V steps, wider than the 9.9V to 10.1V dead-band so duty-cycle dependence is captured across the full operating range. Currents of 0.1, 0.2 and 0.3A are used. At each of the 27 operating points the three-second-averaged $P_{\text{bus,delivered}}$ is recorded against $V_{\text{PSU}} \cdot I_{\text{PSU}}$ and the ratio taken.

V_{bus} (V)	Duty D	η at 0.1 A	η at 0.2 A	η at 0.3 A
7.0	0.58	77.8%	86.9%	89.3%
7.5	0.62	79.0%	87.3%	90.0%
8.1	0.67	81.1%	88.1%	90.4%
8.5	0.71	81.7%	88.4%	90.8%
9.0	0.75	82.1%	88.9%	91.2%
9.5	0.79	82.9%	89.6%	91.7%
10.0	0.83	83.8%	89.9%	92.5%
10.5	0.87	84.2%	90.6%	92.7%
11.0	0.92	84.7%	91.3%	93.6%

Table 5.1: Measured Import SMPS efficiency by bus voltage and load current.

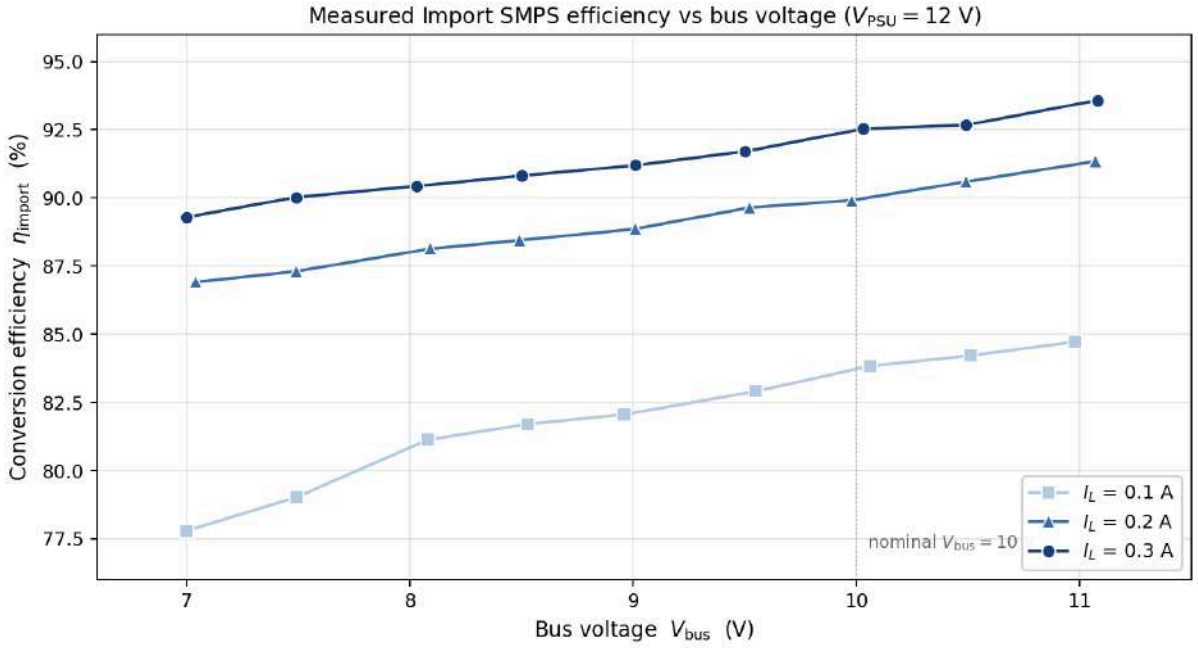


Figure 5.2: Measured Import SMPS efficiency versus bus voltage at three load currents.

At the nominal $V_{\text{bus}} = 10$V the converter delivers 83.8%, 89.9% and 92.5% at 0.1, 0.2 and 0.3A. Both trends (the rise with V_{bus} as duty approaches unity, and the rise with I_L as fixed switching overheads spread over more delivered power) match the buck loss model of Section 2.3 and confirm the converter is operating away from dropout. In cost terms, the 89.9% at nominal 10V and 0.2A inflates the effective import price by $\frac{1}{0.899} \approx 1.11$, an 11% premium the dispatch algorithm of Section 9 sees through its PSU-referenced cost formula.

6 LED Load SMPS

6.1 Design & Simulation

6.1.1 Topology

The LED load is a triple driver assembly: three independent buck channels (red, yellow, blue), each a cut-down SMPS driving a 3W, 1A LED, all under one Pico W. Buck is the natural choice because the 3V LED forward voltage is well below the 10V bus. The hardware differs from the bidirectional modules in two ways the firmware must respect. Current is sensed by an external SPI ADC (MCP3208 family [11]) reading three current channels and three voltage channels through one chip select, with a 0.33 Ω shunt rather than 0.10 Ω, so the inductor current is $i_{\mathrm{L}} = 3 \cdot V_{\mathrm{shunt}}$. Each channel has a dedicated enable pin that must be driven high on every tick; the gate driver gates the PWM off whenever its enable is low. The LED forward voltage is reconstructed from the two ADC reads as $v_{\mathrm{LED}} = 2 \cdot v_{\mathrm{v,pin}} - v_{\mathrm{i,pin}}$, correcting for the sense divider and shunt drop. One of the three identical channels is shown in Figure 6.1.

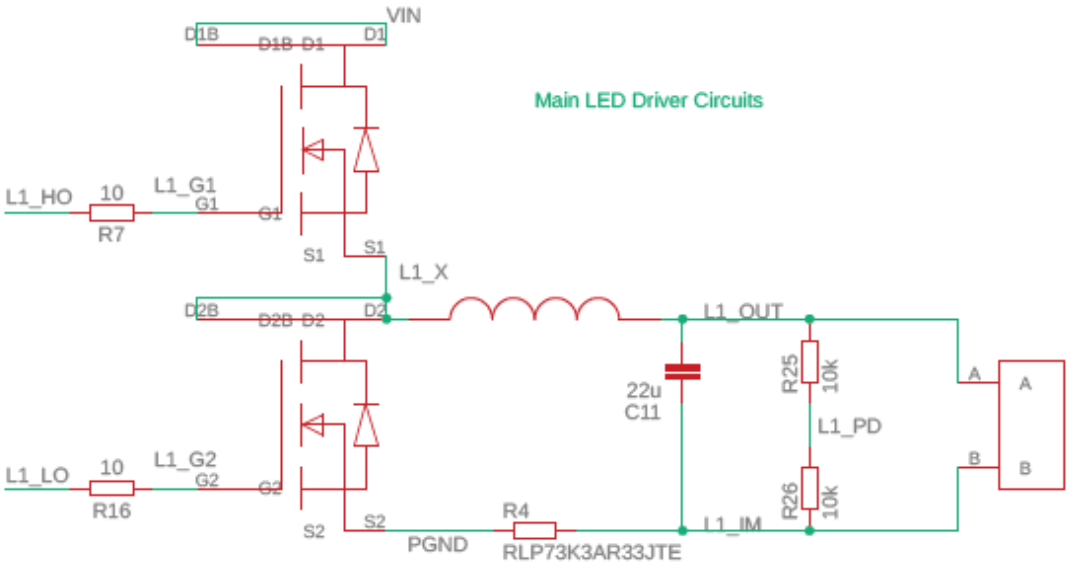


Figure 6.1: LED buck driver power stage, one of three identical channels [12].

6.1.2 Control Mode: CC vs Commanded Power

An LED behaves as a fixed-voltage device, so its power is set by controlling current. The web server hands the system a power figure, so each channel runs commanded-power mode through constant current: power is divided by measured forward voltage to give a current reference $i_{\mathrm{ref}} = \frac{P_{\mathrm{demand}}}{v_{\mathrm{LED}}}$, which a current PI on each channel tracks. The reference is clamped to channel ratings of 3.0W and 1.05A, and a forward-voltage floor guards the division when a string is dark. When demanded power is zero the enable pin is dropped and the PI is reset to stop the integrator winding up while the channel is off.

6.2 Testing & Implementation

The example driver firmware was reworked to drop its external PID dependency in favour of the same inline PI used across the bidirectional modules. The gains and duty saturation range for each channel differ from the bidirectional modules to suit the 0.33 Ω shunt and the driver's

own PWM limits. Each LED has its own protection: if any channel's current exceeds 1.4 times its rated maximum the firmware latches a trip and records which colour tripped, so a faulty LED is diagnosable from the logged readings instead of showing up as a generic fault.

A startup check helped separate wiring faults from real ones. The firmware reads every SPI ADC channel a few times before any PWM runs; if a current reading already looks too high at this point, the cause must be a wiring or measurement problem (a floating ADC input, a wrong port, or a low bus voltage) rather than a real LED overcurrent, and the printed values show which. Shutdown order matters as well: the enable pins are switched off before the duty cycles are zeroed, and since the enable pin cuts the PWM at the driver itself, the LEDs turn off immediately. A 10-second watchdog that switches all channels off when commands stop arriving exists in the code but is currently disabled (Section 8.3); instead, the operator can press Stop on the dashboard, which sends a zero-power command to every LED channel.

6.3 Evaluation & Optimisation

This section reports the power-following accuracy of each channel (which verifies Requirement 1), then explains why conversion efficiency is not separately characterised.

6.3.1 Power-Following Accuracy

A 60-second scripted test exercises each channel individually and then together, with each channel's commanded and measured power logged from the dashboard at 20 Hz. Figure 6.2 overlays the two: the faint trace is commanded power, the dark trace is measured power delivered to the LED string. The sequence is red at 2W for 6s, off, yellow at 2W for 6s, off, blue at 2W for 6s, off, all three together at 1, 2 and 3W for 10s, off, and a 15-second simultaneous ramp from 0 to 3W.

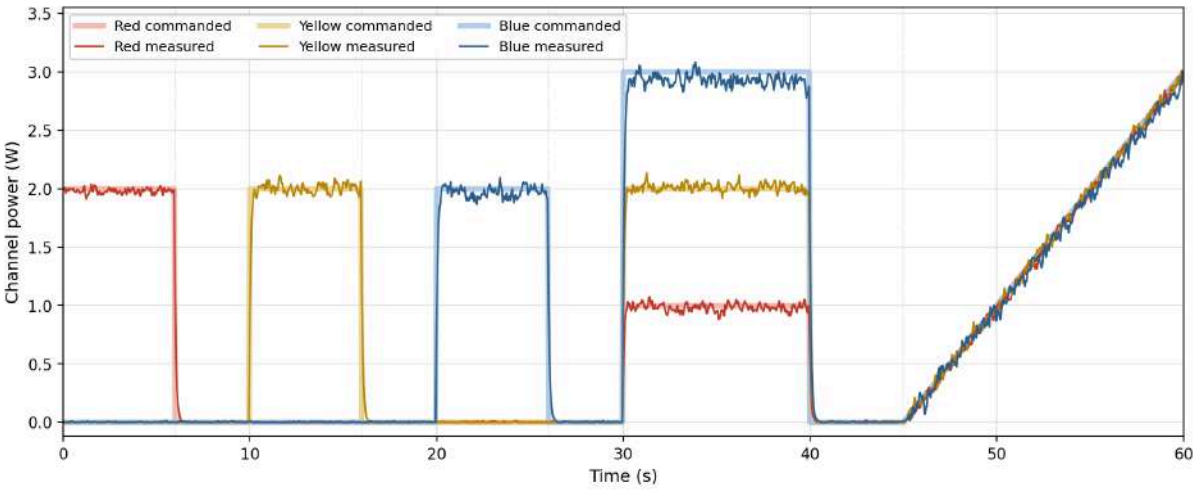


Figure 6.2: Commanded and measured LED power for each channel across the 60-second tracking sequence.

Each channel tracks within about ±3% across all operating points, with the residual driven by forward-voltage variation between strings that the $i_{\text{ref}} = \frac{P_{\text{demand}}}{v_{\text{LED}}}$ division (Section 6.1.2) cannot compensate when v_{LED} is itself estimated through the SPI ADC chain. Step settling on each on/off transition resolves inside one 50 ms dashboard sample. The all-on and ramp phases confirm channels do not interact through the shared bus or SPI ADC, so a multi-colour demand is served additively.

6.3.2 Bus-Side Measurement Gap

The LED Load conversion path sits between the bus meter and the LED string, in the same position in the power path as Export. The lecturer's bus meter registers $P_{\text{LED,bus}} = P_{\text{LED,channel}} + P_{\text{losses}}$ as load, while firmware reports only the post-loss values p_{red}, p_{yel}, p_{blu} measured at the LED string. As a result, the dispatch algorithm reads an LED demand under-reported by exactly the LED Load buck conversion loss, and the same effect applies on Export where the post-loss dissipation under-reports the bus-credited revenue (Section 5.3). The buck loss could in principle be attributed to either side of the converter, but with no instrumentation between the bus and the SMPS we cannot measure where it actually lands. The practical option is to maximise η_{LED}: at $\eta_{\text{LED}} = 1$ the post-loss value would equal the bus value and dispatch decisions would align exactly with the lecturer's reading; at realistic efficiencies the algorithm runs on slightly optimistic numbers and PnL is marginally worse than a perfectly-informed policy would achieve. The proper fix is a dedicated INA219 between the bus and the SMPS, feeding the true bus-side current to the dashboard so dispatch operates on the lecturer's view of the load; this was identified but not implemented within the project window.

7 Measurement Filtering

Noise on the power readings obscured the operating-point comparisons Sections 3.3, 4.2 and 5.3 depend on. Typical oscillation on P_{panel} was several hundred milliwatts peak-to-peak against a steady-state mean of a few watts, with similar relative magnitudes on the other modules. The fix is a three-layer attenuation applied identically across the PV, supercapacitor, import and export modules.

7.1 Hardware Averaging

The bidirectional template inherits an INA219 CONFIG register of 0x199F (12-bit, single-sample, 532 µs conversion [13]), which lets most of the switching ripple through unaveraged on the current-sense shunt. Reprogramming to 0x1E67 (16-sample averaging, 8.51 ms conversion) across all four modules clears this, in which the longer conversion time is still over twenty times faster than the slowest downstream consumer, the supercapacitor's 200 ms OCV window (Section 4.2). PV and supercapacitor pick this up through identical edits to their inline `ina_init` functions, and import and export through a single edit to the shared `INA219` class.

Voltage channels are read by the Pico's onboard 12-bit ADC [14] through a resistive divider and vary by several LSBs sample-to-sample regardless of INA219 settings. Each `read_va` and `read_vb` call performs four single reads and right-shifts the sum by two to take the mean:

$$V_{\text{oversampled}} = \frac{V[0] + V[1] + V[2] + V[3]}{4}$$

Cost per call is about 10 µs, well inside the 1 ms control tick budget. Peak-to-peak jitter falls by roughly 1.6×, with diminishing returns past four samples.

7.2 Exponential Moving Average

A first-order EMA is applied in software on top of the hardware-averaged readings:

$$V_{\text{filt}}[n] = (1 - \alpha) \cdot V_{\text{filt}}[n-1] + \alpha \cdot V[n]$$

with $\alpha = 0.05$ at the 1 kHz sample rate, giving a time constant $\tau \approx 20$ ms and a corner frequency near 8 Hz. The filtered signals drive the dashboard display, the three-second block averages, the P&O power comparison (Section 3.2) and the cycle-test energy integrators (Section 4.2). $\alpha = 0.05$ is bounded both ways: too small (longer τ) and the filter adds phase lag that slows displayed transients on dashboard reads; too large (shorter τ) and it passes switching-ripple aliasing through. 20 ms gives about 60 ms to 95% step settling, which fits inside the 150 ms second-half averaging window of the P&O dwell and the 200 ms tail of the cycle-test settle phases. Figure 7.1 shows the resulting magnitude and step response.

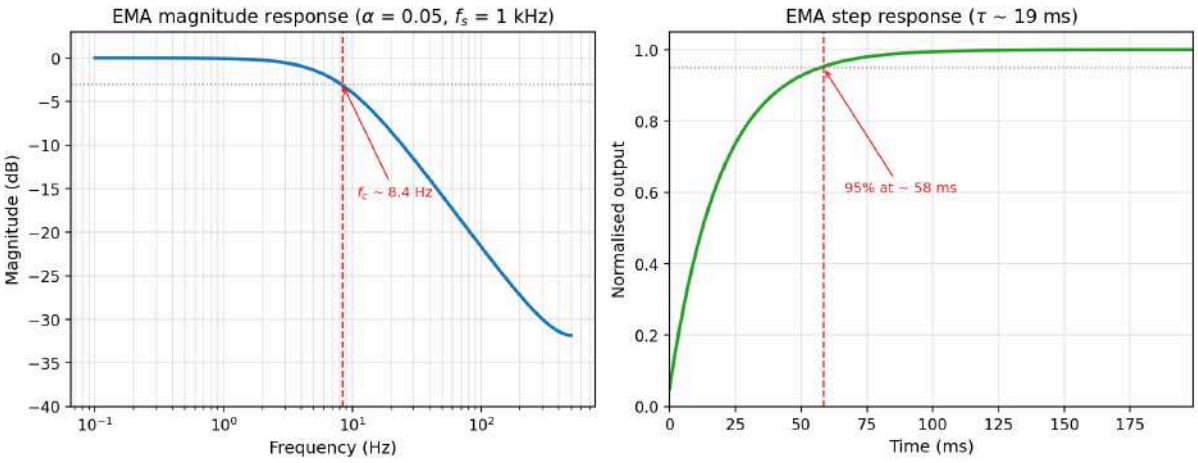


Figure 7.1: Measurement EMA magnitude and step response at alpha = 0.05.

The inner PI loops and safety guards bypass the software EMA and block-average outputs, mainly for two reasons. First, adding 20 ms of phase lag to a current PI tuned against unfiltered inputs would shift its stability margin and force a retune. Second, the trip thresholds (panel undervolt, bus overvolt, overcurrent) were designed for instantaneous response; a filtered safety guard would react 20 ms late, allowing more energy into a fault before tripping. Keeping the filter on the display and characterisation paths only preserves control-loop behaviour and trip response while still giving the operator stable readings.

7.3 Three-Second Block Average

Each module computes a true block average of its bus-side power over a three-second window in firmware:

$$P_{\text{avg3s}} = \left(\frac{1}{N}\right) \sum_{i=1}^{N} P[i]$$

with $N = 3000$ at 1 kHz. The window restarts on changes to the commanded operating point (MPPT mode for PV, I_{cmd} for supercapacitor, V_{target} and I_{max} for import and export), so each completed average reflects one operating point. Rounding the operating-point values before comparison avoids spurious resets from tiny floating-point differences introduced by JSON parsing. The computed average lives in each module's data dictionary; persistence into the backend SQLite store and rendering on the operator dashboard are scheduled wiring work, so the side-by-side comparisons in Sections 3.3 and 5.3 read the filtered live readings directly. Inside the supercapacitor cycle test the block average sits on top of phase changes as a real-time sanity check that the commanded current has settled.

7.4 Evaluation

The three-layer treatment is evaluated by the noise reduction it delivers on each module's bus-side power readings. Figure 7.2 overlays a raw single-sample trace against the same operating point after the three-layer filter, shifted to a 2W PV operating point for visual context.

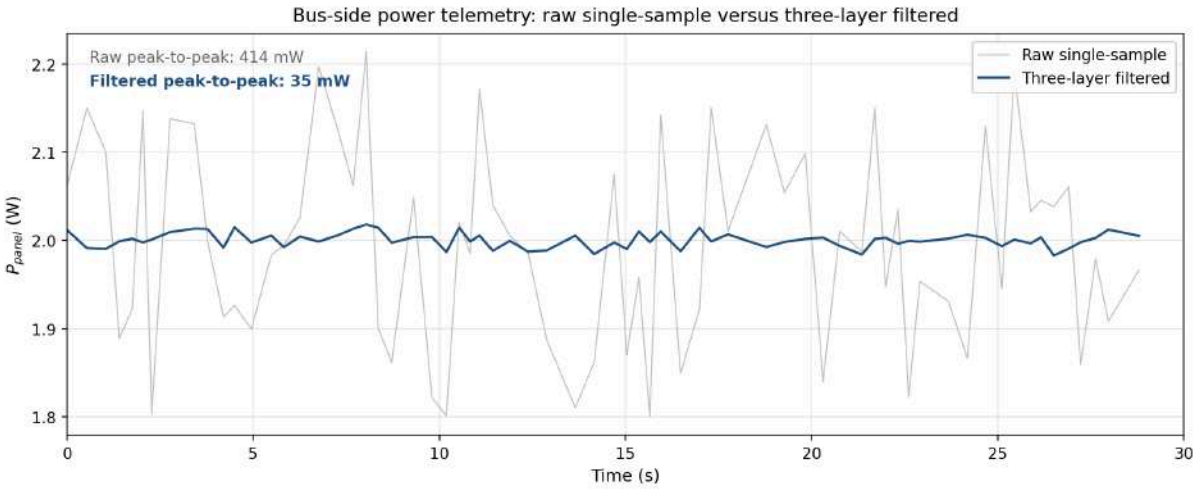


Figure 7.2: Bus-side power readings: raw single-sample versus three-layer filtered.

Raw P_{panel} oscillation falls from about 300 mW peak-to-peak at the 1 kHz tick to under 10 mW per P&O dwell-averaged comparison (Section 3.2), roughly a 30× reduction. Equivalent attenuation on the supercapacitor, import and export modules brings the three-second-averaged readings inside a few milliwatts, fine enough to resolve the 1.4 percentage-point irradiance-to-irradiance spread in the PV efficiency sweep (Section 3.3) and the 1 to 2 percentage-point steps between adjacent supercapacitor charge currents in the round-trip sweep (Section 4.2).

8 Software System

The software system links the Azure simulation server, backend services, embedded power modules and operator dashboard into one communication, control and monitoring platform.

8.1 Architecture

The lowest tier is the firmware for the five Pico-controlled power modules: PV, grid import, export, supercapacitor and LED load. Each runs a lightweight HTTP server on core 1 and exposes a JSON `POST /cmd` endpoint. Separating WiFi and socket handling from the 1 kHz control loop on core 0 reduces control jitter. The networking thread also uploads telemetry through role-specific `POST /modules/tlm/{role}` requests. A 1 s heartbeat bounds normal monitoring delay and limits WiFi and database load; a command-trace change triggers an additional push after the new setpoint has been applied.

The middle tier is a Python FastAPI backend with a SQLite store. In normal operation it is the bridge between browser clients and modules: FastAPI forwards commands to the addressed Pico, while pushed telemetry is timestamped, persisted and served as latest or historical snapshots. A standalone poller ingests Azure data and stores at most one grid snapshot per `(day, tick)`, enforced both in code and by a database uniqueness constraint.

The top tier is a React single-page application built with Vite for browser-based monitoring, visualisation and control.

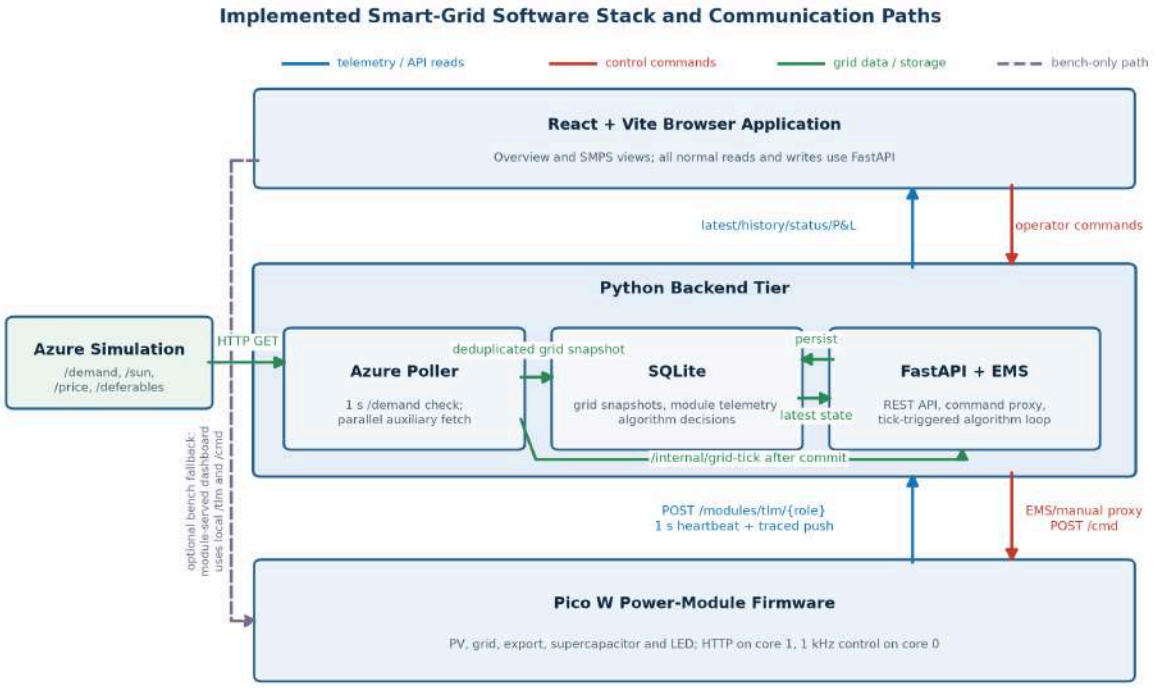


Figure 8.1: Software stack showing module readings flowing upward to FastAPI and control commands flowing downward through the same backend.

As summarised in Figure 8.1, telemetry is pushed to `/modules/tlm/{role}`. Operator and EMS commands use `/modules/cmd/{role}`, after which FastAPI forwards the JSON payload to the module's `/cmd`.

8.2 Backend and Control Layer

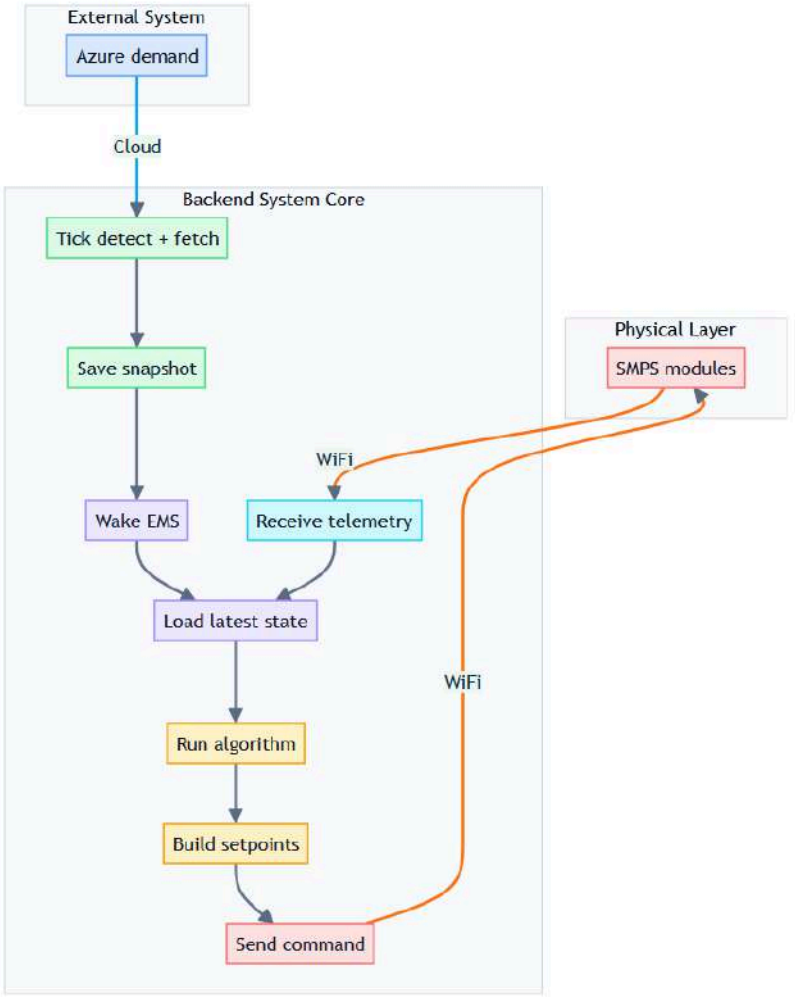


Figure 8.2: EMS control flow from Azure tick ingestion through FastAPI-mediated command and telemetry paths.

The backend is the central coordination layer. It stores telemetry and grid snapshots and hosts the Energy Management System (EMS) loop, which reads the latest committed state, runs the selected algorithm and dispatches setpoints to the LED, supercapacitor, grid and export modules. The PV converter operates through its own MPPT control rather than EMS dispatch.

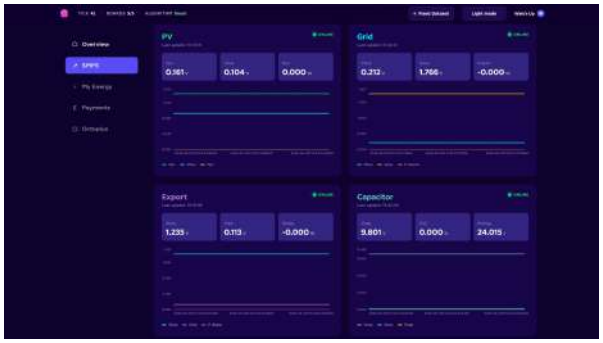
The Azure poller nominally checks `/demand` once per second and uses `(day, tick)` as the change detector. On a new tick it fetches `/sun`, `/price` and `/deferables` in parallel, commits a snapshot, then calls `/internal/grid-tick`. The EMS loop wakes on this event, with a 5 s database fallback, and does not reapply the same tick. It skips dispatch if the grid snapshot is older than 15 s or if any of the four actuator modules reports a trip. The dashboard's online status is separate: a module is offline when its latest telemetry is more than 6 s old.

Algorithms are registered and switchable at runtime. Applied EMS decisions include the algorithm identifier, inputs, recommendation and the command results for each module; together with grid and telemetry snapshots, these support replay and offline policy comparison. Manual dashboard commands use the same proxy but are not separately written to the decision table.

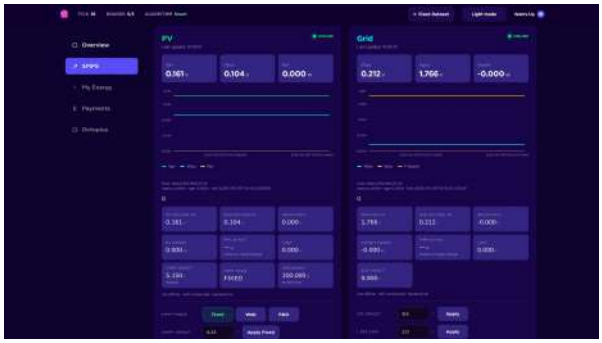
Run utilities provide auto-run start/stop, one-tick testing, algorithm selection, CSV export and run-scoped import-cost, export-revenue and net-cost summaries. Stopping auto-run sends explicit safe defaults to all four actuators.

8.3 Frontend Dashboard

(a) Overview: algorithm state, module health and grid trends.



(b) SMPS system monitoring.



(c) Advanced algorithm/board decision view.

Figure 8.3: Operator interface showing the high-level overview and focused SMPS monitoring and algorithm-control panels.

The frontend uses a responsive card-based layout to divide system state into clear functional groups. Summary cards expose the most important values at a glance, while charts and module cards provide progressively more detail.

The default **Overview** page is for users monitoring the whole smart-grid system rather than individual converter details. It summarises demand, solar availability, energy prices, deferrable loads, active algorithm, run P&L and the health of all five power modules. Recent demand and irradiance charts show how the operating conditions evolve over time.

The **SMPS** page is the engineering view of the control system. It brings the active EMS algorithm, bus-level energy flow and measurements from each Pico-controlled module into one page, allowing the interaction between the algorithm and physical converters to be observed. Its **Advanced** mode adds detailed module diagnostics, manual setpoint and protection controls, algorithm selection and EMS execution controls. This mode is therefore used for board bring-up, fault investigation and controlled manual testing, while normal system operation remains accessible through the simpler Overview and SMPS monitoring cards.

8.4 Design Trade-offs

The first Azure poller fetched the complete state every five seconds. Because a sequential cycle could exceed the simulator update period, ticks could be missed. The final design uses the smaller `/demand` response for nominal 1 s change detection and fetches the three slower endpoints in parallel only after a change. This reduces detection delay and unnecessary traffic, although network request time means a strict sub-second bound is not guaranteed.

Early versions also polled each module's `/tlm` endpoint. Offline boards caused repeated timeouts and scaling cost, so the final design makes modules push telemetry. This removes backend polling waits, but freshness now depends on each board's WiFi uplink.

Commands remain backend-mediated so the EMS can run without an open browser and automated and manual control share one addressing and communication layer. This also avoids browser-to-Pico CORS and network-routing problems. The trade-off is that FastAPI is a control-path dependency during normal coordinated operation.

8.5 On-Module Fallback Dashboard

For bench bring-up, each power-module HTTP server can serve a single-file `dashboard.html` from the Pico filesystem, reading `/tlm` and writing `/cmd` for local single-module control when FastAPI is unavailable. Telemetry logging, EMS automation and multi-module coordination still require the backend path.

9 Algorithm Software

9.1 Coordinator Loop and Economic Objective

The coordinator is a background thread inside the FastAPI backend, woken on each new Azure tick (the loop mechanics, including deduplication and the 15-second and trip skips, are detailed in Section 8.2). On each pass it reads the latest grid, PV and supercapacitor state, evaluates the active algorithm's `decide(state)` function, and dispatches commands to the four actuators (LED Load, supercapacitor, grid, export) in that fixed order. Because each module's inner loop holds its last setpoint autonomously, a coordinator stall cannot destabilise the bus: droop, PI control and each module's clamps run independently at 1 kHz.

The objective is to minimise the net cost of grid energy over a daily Azure cycle while always meeting demand. Following the metering setup of Section 2.3, import is debited at the PSU reading and export credited at the bus reading, so the daily cost is

$$C = \sum_t \big(P_{\mathrm{PSU}}(t) \cdot p_{\mathrm{sell}}(t) - P_{\mathrm{export,bus}}(t) \cdot p_{\mathrm{buy}}(t)\big) \cdot \Delta t$$

with $P_{\mathrm{PSU}}(t) = P_{\mathrm{import,bus}} \frac{t}{\eta_{\mathrm{import}}}(V_{\mathrm{bus}}, I_L)$ coupling the import draw to the efficiency map of Section 5.3. Prices follow the grid's perspective: p_{sell} is what the utility charges us on imports, p_{buy} what it credits us on exports, with $p_{\mathrm{buy}} \approx 0.5 \cdot p_{\mathrm{sell}}$ at every tick. That spread is the main opportunity the algorithm can exploit. The coordinator's available actions are deferrable scheduling, supercapacitor charge or discharge, and net import or export, subject to instantaneous demand being met, each deferrable receiving its full energy inside its window, and the supercapacitor and grid modules staying inside their hardware envelopes.

9.2 Naive Baseline

The naive policy is a five-step reactive symmetric-droop rule. PV is first derated to bus side by a conservative $\eta_{\mathrm{PV}} = 0.90$ (the measured value in Section 3.3 is 96 to 98%, so the bias keeps the algorithm on the surplus side of modelling error). Active deferrables are then spread linearly across their remaining real-time window so the rate is $\frac{E_{\mathrm{remaining}}}{5 \cdot N_{\mathrm{ticks\ remaining}}}$ watts each tick, with adaptive catch-up baked in: under-serving last tick raises the next tick's rate. Effective load is base demand plus the sum of those rates and is commanded to the LED Load, split three ways across red, yellow and blue channels with each firmware-clamped to 3 W. The net imbalance $P_{\mathrm{net}} = P_{\mathrm{PV,bus}} - P_{\mathrm{load,eff}}$ is then offered to the supercapacitor through a 1 W deadband (noise floor of the bus), with charge current $\frac{P_{\mathrm{net}}}{V_{\mathrm{bus}}}$ clamped to 0.6 A and refused above 15.7 V, discharge current $|P_{\mathrm{net}}| \frac{}{V_{\mathrm{bus}} \cdot \eta_{\mathrm{cap}}}$ refused below 10.5 V and linearly tapered through the 10.5 to 11 V firmware soft region. Anything the supercapacitor cannot absorb spills to droop: the grid and export modules sit permanently enabled at 9.9 V and 10.1 V respectively and passively make up the residual.

Figure 9.1 shows the naive policy across one Azure day on the grid. The cap power panel is small and twitchy, charging only when noise pushes the bus above the deadband; the grid panel does almost all of the work, importing through the morning, evening and night and exporting only the small midday surplus that the cap could not absorb.

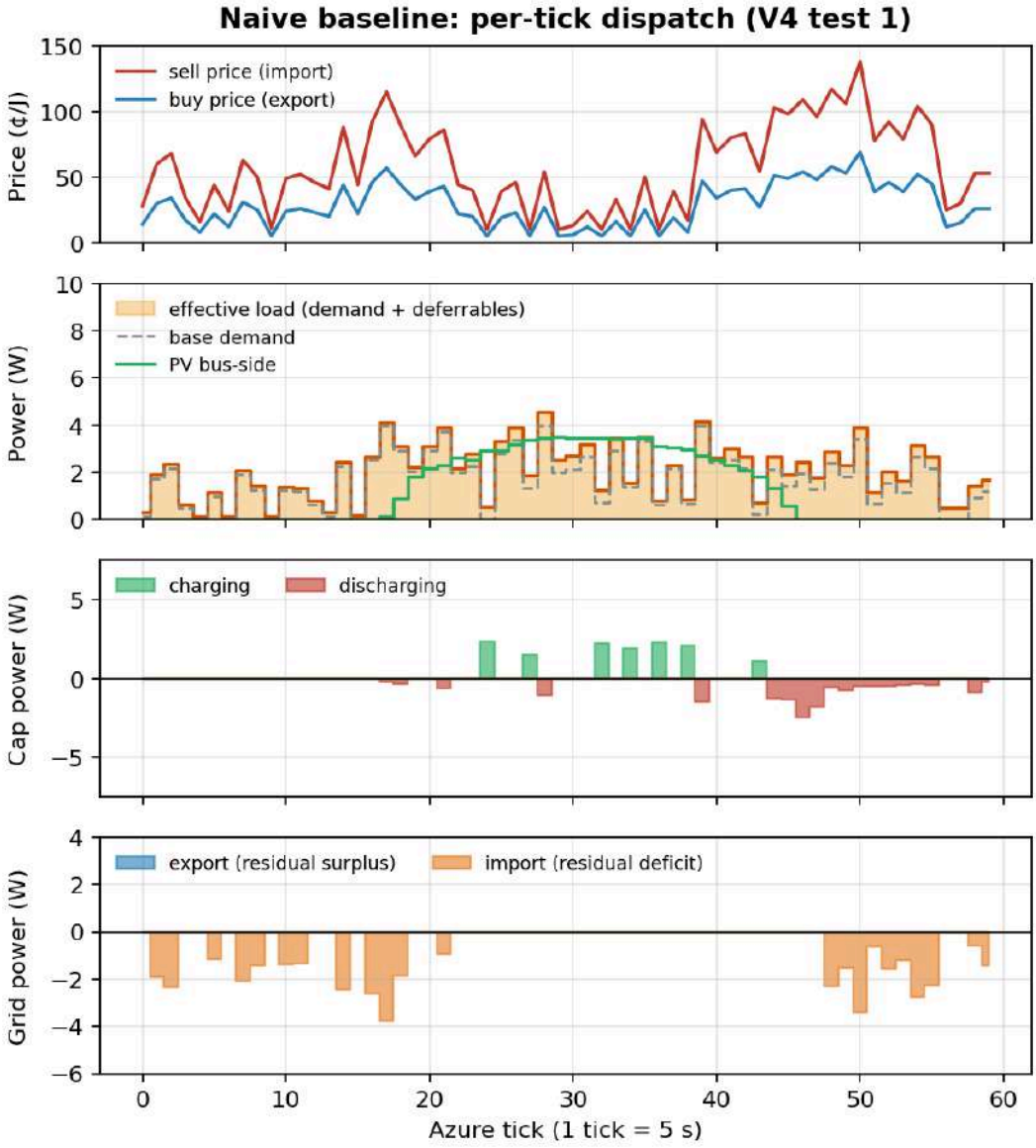


Figure 9.1: Naive baseline: tick-by-tick dispatch over one Azure day (V4 test 1).

9.3 Smart V1: Forecast-Aware Scheduling and Arbitrage

Smart V1 keeps every safety guard of the naive policy (LED clamp, supercapacitor taper, 0.6 A clamp, deadband, discharge-side η_{cap} compensation) and replaces two of its reactive steps with forecast-aware ones. The forecast itself is closed-form: the Azure simulation has a deterministic mean (sun follows a sine over ticks 15 to 45, base demand follows a piecewise-linear envelope, sell price = max(10, 10 + base − sun)), with Gaussian noise on top. Smart V1 evaluates that mean per tick rather than fitting historical data.

The first replacement is the deferrable scheduler. Instead of spreading each deferrable across its window, future ticks are ranked by marginal cost-to-serve: a tick in forecast PV surplus costs p_{buy} (the foregone export revenue), a tick in deficit costs p_{sell} (the imported energy). The scheduler fills the cheapest ticks first, respecting the LED's 9 W capacity each tick and giving the tightest-deadline deferrable first pick. The second replacement is the supercapacitor dispatcher. At every tick it compares the current sell price against the best future sell price across the rest of the day, after subtracting expected leakage, using a piecewise leakage model

fitted to Section 4.2.3 (a 12% dielectric step over the first minute, then 0.25% per Azure tick). It picks one of six actions: an end-of-day drain past tick 55, a discharge when the current price beats the best future price by more than 20 ¢/J, a charge when the forecast return exceeds the charging cost by the same margin, an import-and-charge action that buys grid energy when a strong forecast peak exceeds the current price by more than 50 ¢/J, an empty-cap refill rule that lowers the import bar when a forecast peak is within eight ticks, and a fall-back passive rule that mirrors the naive deadband when none of these triggers fire. Two pre-peak guards (refuse passive discharge below 70 ¢/J after tick 42, and a fixed price floor instead of forecast comparison) were added after a lab trace showed the supercapacitor being drained at moderate prices and arriving empty at the real evening peak.

Figure 9.2 shows the smart policy on the same day. The supercapacitor panel now has a large midday charge burst (ticks 24 to 32, where PV surplus and cheap prices coincide) and an evening discharge burst (ticks 45 to 50, where the sell price rises above 100 ¢/J). The grid panel imports most of its energy in the cheap morning and exports the midday surplus that exceeds the LED clamp.

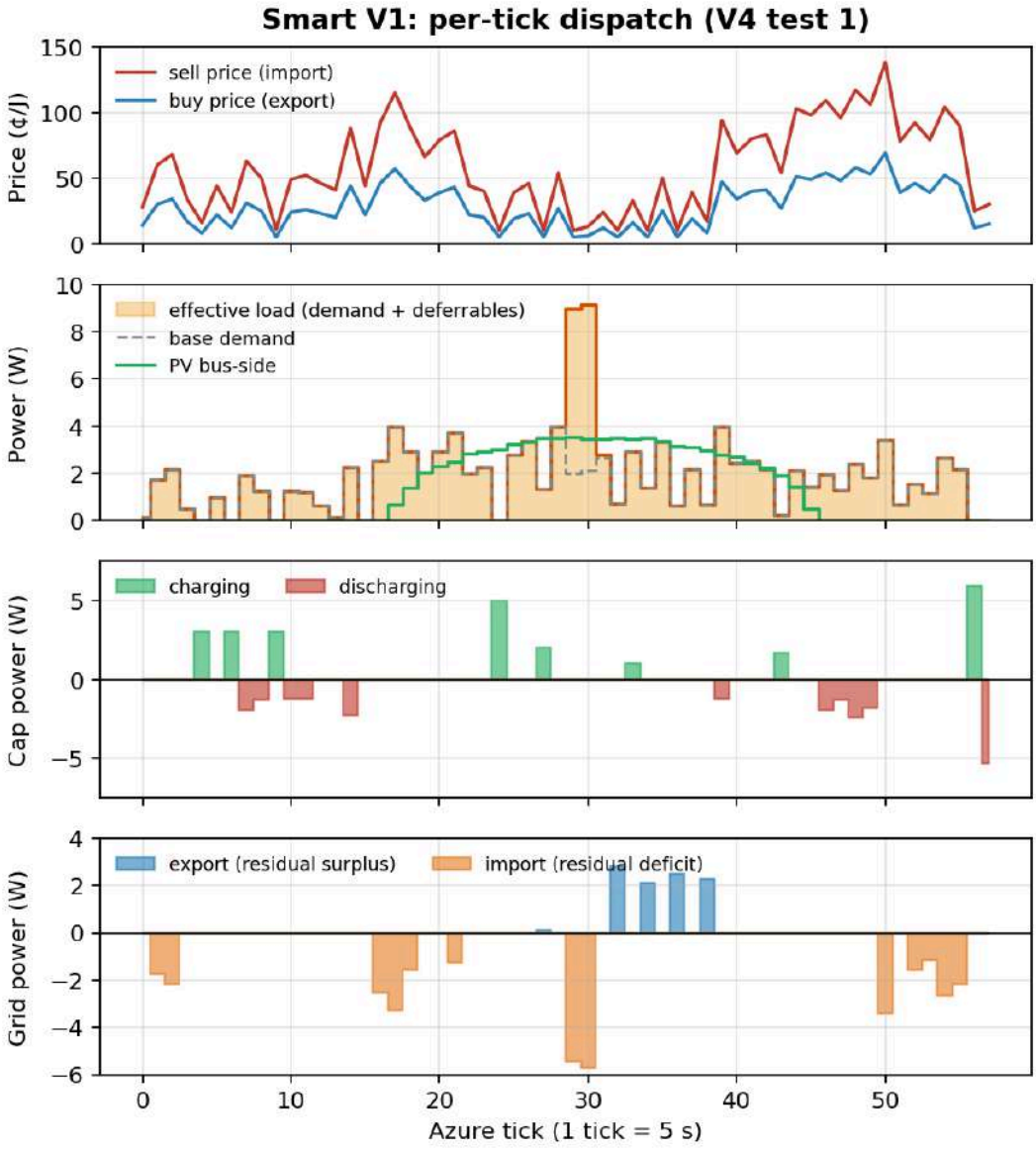


Figure 9.2: Smart V1: tick-by-tick dispatch over the same Azure day (V4 test 1).

9.4 Evaluation

Each policy was run on the grid against the same fixed price, demand and irradiance profile (514 J base demand, 100.6 J of deferrables, 615 J LED target). Net cost integrates the grid module's `p_import` and the export module's `p_dissip` trapezoidally over every consecutive sample pair, multiplied by the sell or buy price in effect at the earlier timestamp; sample gaps longer than 30 s are skipped. The same integrator drives the dashboard's live `/algo/run/cost` P&L card, so Table 9.1 matches what the operator sees.

Run	Import (¢)	Export (¢)	Net (¢)	Saving vs naive
Naive, test 1	34,789	70	34,719	baseline
Naive, test 2	33,493	63	33,430	baseline
Smart, test 1	27,863	450	27,413	21.0%
Smart, test 2	27,447	384	27,064	19.0%

Table 9.1: V4 lab cost summary (cents). Smart V1 saves 20% on average.

Figure 9.3 attributes the saving to where each policy spent its import allowance. Smart V1 imports roughly 3.6 times as much energy in the cheapest band (below 30 ¢/J) and correspondingly less in the 30 to 100 ¢/J middle bands, which is where the naive policy is forced to draw whenever the bus sags. Total imported energy is almost the same across both policies (476 J for smart versus 463 J for naive, averaged across the two paired tests): the saving comes from when the energy is bought, not how much. The supercapacitor cycles about 2.2 times as hard under smart (120 J charged, 98 J discharged on average, versus 54 J and 49 J under naive) and the grid exports about five times as much (36 J versus 7 J), this export being midday PV surplus beyond the 9 W LED clamp while the cap is also charging. Most of the saving comes from price arbitrage on the supercapacitor: cheap-band imports store energy in the cap, which is discharged later to displace what would otherwise be 70 to 100 ¢/J imports.

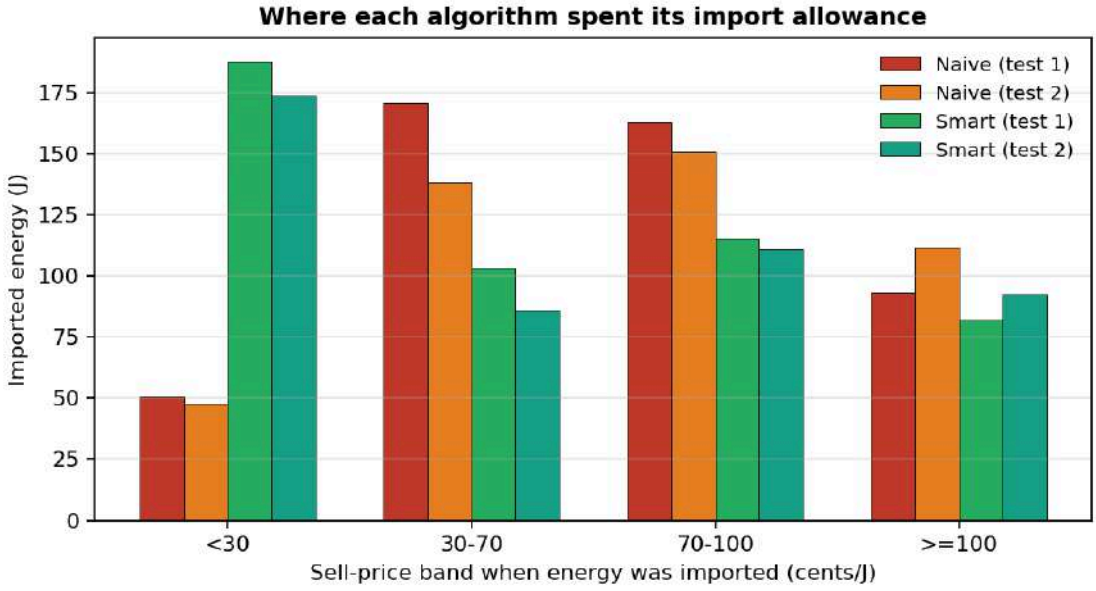


Figure 9.3: Imported energy by sell-price band: smart shifts imports into the cheap band.

9.5 Backtest on 1000 Simulated Days

The two lab runs sample a single scripted day, so both policies were also replayed offline over the 1000 randomly generated days in `datasets.json` to check that the saving holds across price profiles. The replay applies the same physical model as the firmware: PV follows the deterministic sun profile, the supercapacitor carries 90% SMPS efficiency in each direction with 0.25% leakage per tick, the LED load is clamped at 9W, droop closes the bus balance, and any deferrable energy left unserved at the end of a day is charged at that day's mean sell price.

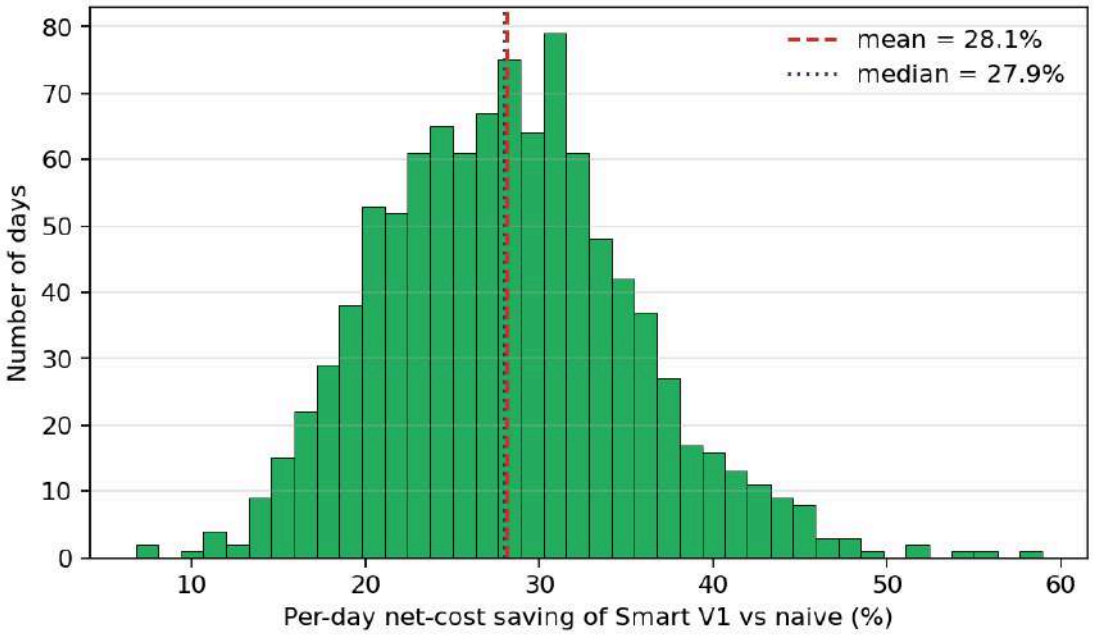


Figure 9.4: Distribution of daily saving of Smart V1 over naive across 1000 days.

Metric	Value
Mean daily saving	28.1%
Median daily saving	27.9%
Standard deviation	7.3%
Worst day / best day	6.8% / 58.9%
5th to 95th percentile	16.9% to 41.0%
Days where Smart V1 wins	1000 of 1000
Aggregate cost saving	27.9%

Table 9.2: Smart V1 vs naive backtest summary over 1000 simulated days.

Figure 9.4 shows the distribution never crosses zero: Smart V1 beats the naive policy on all 1000 days, the worst still 6.8% cheaper. The backtest mean sits above the 20% measured in the lab because the simulation has no command latency and no measurement noise, both of which blunt the timing of charge and discharge decisions on the real grid. The spread reflects how much each day's price profile offers: flat days leave little for the forecast to exploit, while a deep midday trough followed by a sharp evening peak rewards it most.

9.6 Auxiliary Power Overhead

Figure 9.5: USB inline meter: a single Pico W draws 0.262 W on its 5 V rail.

The five Pico W controllers run from a separate USB powered hub, unmetered by the algorithm. A USB inline meter reads 0.262 W per module (Figure 9.5), so the aggregate fixed draw is 1.31 W or 393 J per 300 s Azure day. Total energy throughput across the grid (PV panel-side, supercapacitor charge and discharge, grid import and export, LED dissipation) averaged 1724 J per day, or 5.75 W. The controllers therefore account for $\frac{1.31}{1.31+5.75} \approx 19\%$ of total grid power. However, this is a fixed cost: highly inefficient on a small-scale grid, but a larger supercapacitor and higher demand would improve it.

10 Mechanical Design

10.1 3D-Modelled SMPS Enclosure

The four bidirectional SMPS boards sit in a 3D-printed chassis that protects the exposed power electronics from accidental shorts and keeps the assembly stable enough to move and demo without disturbing the wiring.

The v1 print (Figure 10.1, left) had solid side walls and a solid base. It was mechanically fine but trapped heat: with no intake from the sides or below, the lower boards had no convection path.

The v2 revision (Figure 10.1, right) added vertical slots in the side walls and vents in the base, so cool air enters from below and rises through to the open top. 3 mm standoff ribs on the inside walls hold each board off the wall so air flows between them.

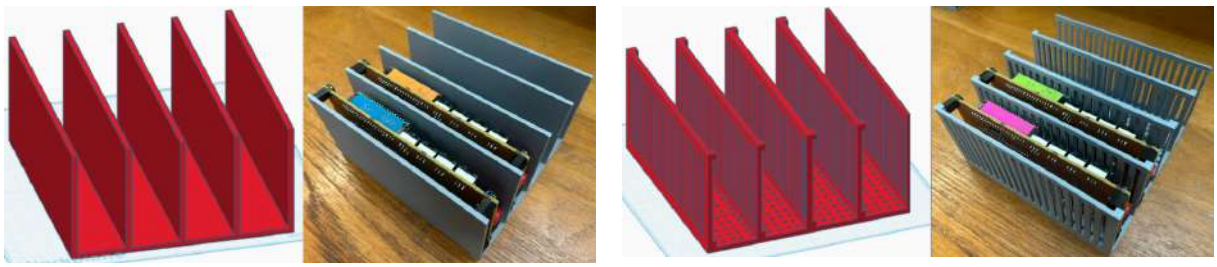


Figure 10.1: SMPS stand iteration: v1 (left) and v2 (right)

11 Extra Test & Requirement Verification

11.1 Communication & Integration Tests

Figure 11.1: Median processing and communication latency measurements across 20 confirmed Smart V1 traces.

Twenty Smart V1 traces were captured during autonomous operation. Each command carried a trace identifier returned in telemetry, linking an Azure update across state acquisition, algorithm execution, HTTP dispatch and Pico control-loop adoption. The median time from detecting a new Azure response to receiving confirming supercapacitor telemetry was 1417 ms; including the initial `/demand` request, the median path was 1759 ms. This indicates that remaining delay was dominated by communication and data handling rather than algorithm execution, identifying the communication path as the main target for future optimisation.

The test confirms autonomous operation across the Azure-to-backend-to-SMPS path. Further improvement should focus on persistent connections or a lighter messaging protocol, event-driven socket handling and a controlled wireless access point. The present confirmation measurement follows the supercapacitor command path; the trace method can be extended to other actuators and network conditions.

11.2 Requirements Verification

Each of the 6 project requirements [10] is mapped below to the subsystem and test that verify it.

Req	Requirement	Realised by	Verification	Status
1	Supply domestic LEDs per web demand	LED Load SMPS (Section 6)	Power-tracking test (Section 6.3)	Implemented
2	Extract PV energy with MPPT	PV SMPS (Section 3)	MPPT mode comparison	Implemented
3	Store excess in supercapacitor	Supercapacitor SMPS (Section 4)	Cycle test; round-trip efficiency	Implemented
4	Import and export to grid, metered	Import/Export SMPS (Section 5)	Import efficiency sweep	Implemented
5	Minimise cost and beat naive	Algorithm (Section 9)	Dry run versus naive baseline	Implemented; 20% improvement
6	UI for current and historic energy data	User Interface (Section 8)	Live dashboard; history plots	Implemented

Table 11.1: Brief requirements: verification method and current status.

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